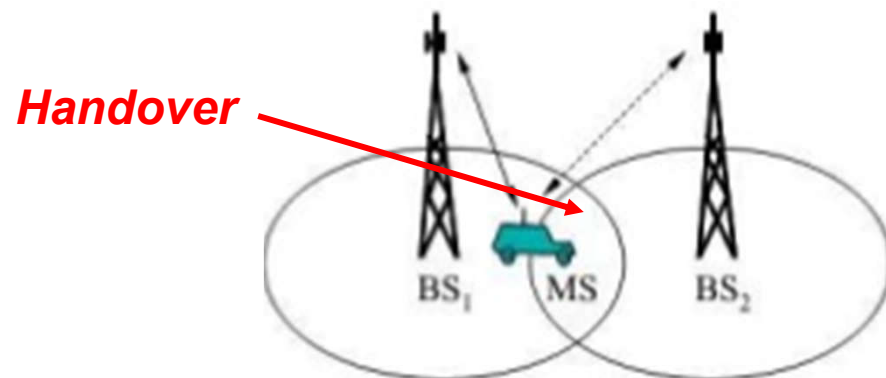


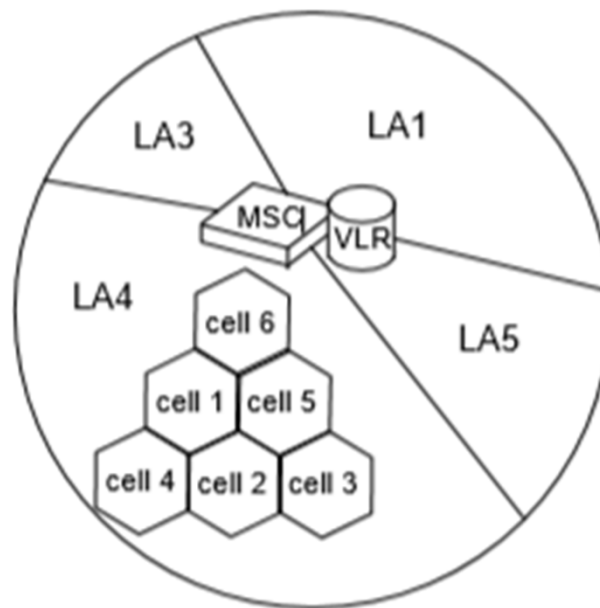
II. GSM network architecture (cont.)

- ▶ A Location Area (LA) is defined as a group of cells. Within the network, a subscriber's location is known by the LA which they are in. The identity of the LA in which an MS is currently located is stored in the VLR
- ▶ When an MS crosses a boundary from a cell belonging to one LA into a cell belonging to another LA, it must report its new location to the network. When an MS crosses a cell boundary within an LA, it does not need to report its new location to the network. When there is call for an MS, a paging message is broadcast within all cells belonging to an LA.



II. GSM network architecture (cont.)

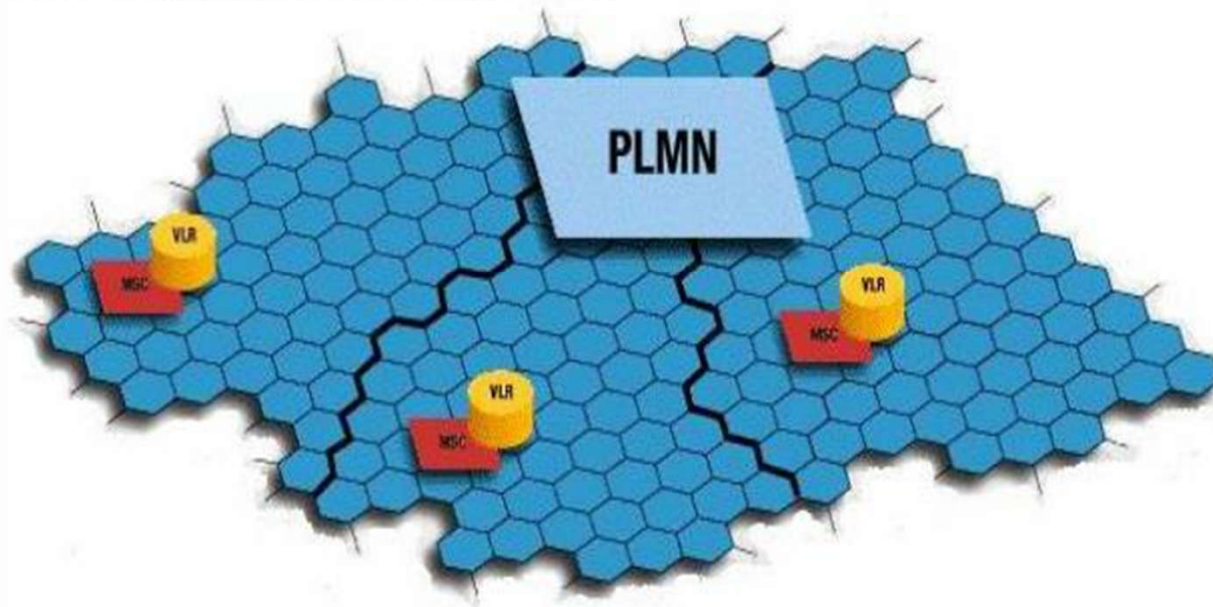
- ▶ An **MSC service area** is made up of a number of LAs and represents the geographical part of the network controlled by one MSC. In order to be able to route a call to an MS, the subscriber's MSC service area is also recorded and monitored. The subscriber's MSC service area is stored in the HLR



II. GSM network architecture (cont.)

- ▶ A Public Land Mobile Network (PLMN) service area is the entire set of cells served by one network operator and is defined as the area in which an operator offers radio coverage and access to its network. In any one country there may be several PLMN service areas, one for each mobile operator's network.

Public Land Mobile Network (PLMN)



II. GSM network architecture (cont.)

An MS can have one of the following states:

- ▶ **Idle:** the MS is ON but a call is not in progress
- ▶ **Active:** the MS is ON and a call is in progress
- ▶ **Detached:** the MS is OFF

Mode	Term: Description
Idle	Registration: the process in which an MS informs a network that it is attached
	Roaming: When an MS moves around a network in idle mode, it is referred to as roaming
	International Roaming: When an MS moves from in PLMN of a country to another PLMN of other country
	Location Updating: An MS roaming around the network must inform the network when it enters a new LA.
	Paging: This is the process whereby a network attempts to contact a particular MS. This is achieved by broadcasting a paging message containing the identity of that MS.
Active	Handover: This is the process in which control of a call is passed from one cell to another while the MS moves between cells

III. Wireless Transmission

1. Some main characteristics

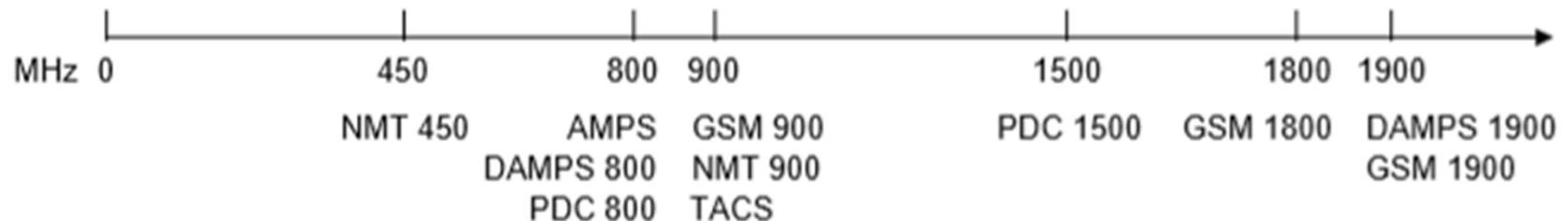
► Frequency characteristics:

System	P-GSM 900	E-GSM 900	GSM 1800	GSM 1900
Frequencies: • Uplink • Downlink	890-915 MHz 935-960 MHz	880-915 MHz 925-960 MHz	1710-1785 MHz 1805-1880 MHz	1850-1910 MHz 1930-1990 MHz
Wavelength	~ 33 cm	~ 33 cm	~ 17 cm	~ 16 cm
Bandwidth	25 MHz	35 MHz	75 MHz	60 MHz
Duplex Distance	45 MHz	45 MHz	95 MHz	80 MHz
Carrier Separation	200 kHz	200 kHz	200 kHz	200 kHz
Radio Channels ¹	125	175	375	300
Transmission Rate	270 kbits/s	270 kbits/s	270 kbits/s	270 kbits/s

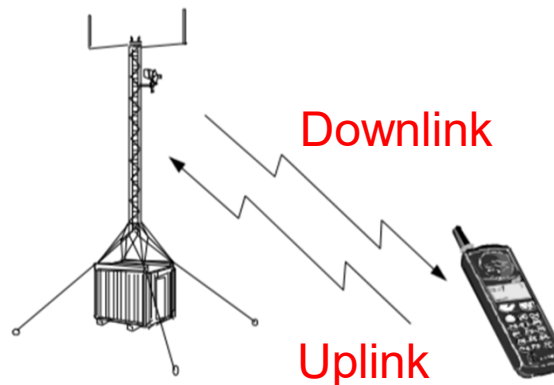
III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

- Frequency standards:



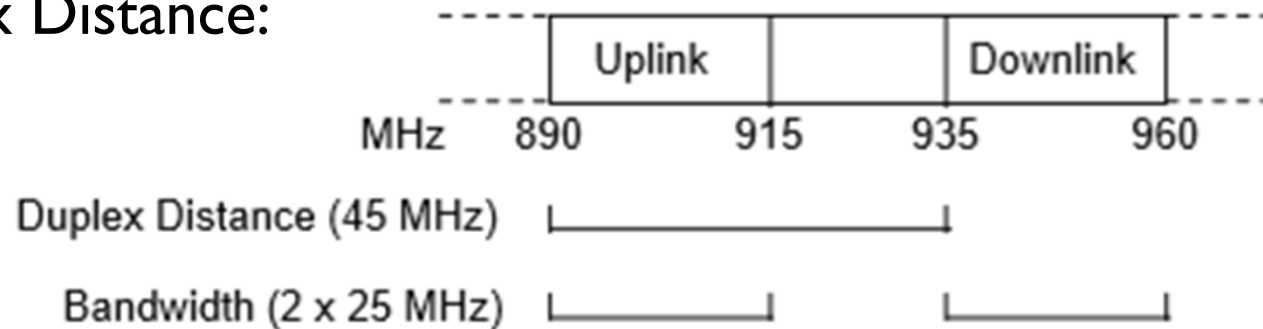
- Channel: A duplex channel, two frequencies. One for the direction from the MS to the network called **uplink** and the other from the network to the MS called **downlink**



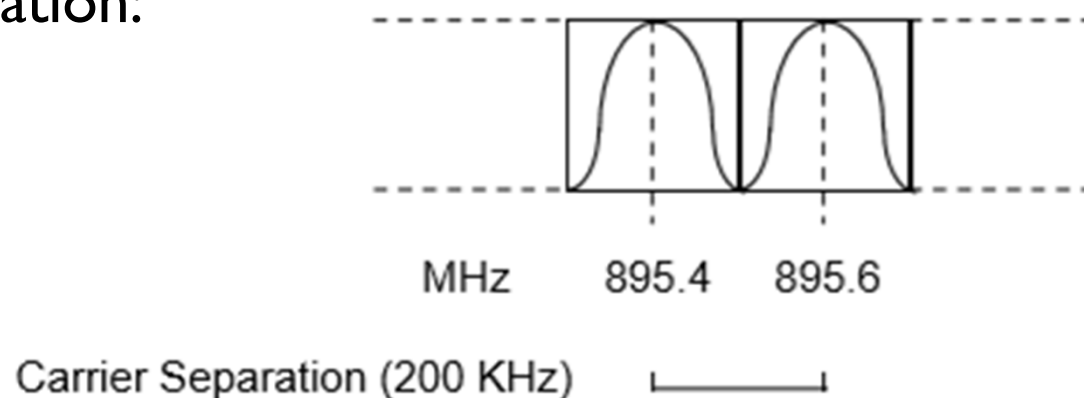
III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

► Duplex Distance:



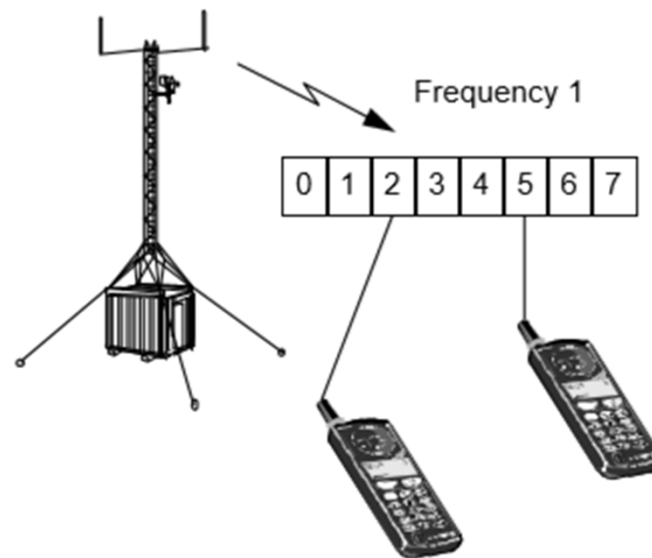
► Carrier Separation:



III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

- ▶ Modulation method: Gaussian Minimum Shift Keying (GMSK).
- ▶ Access method: Time division multiple access (TDMA)



III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

► Problems about frequencies:

- ▼ It is the number of frequencies in a cell which determines the cell's capacity.
- ▼ Each company with a license to operate a mobile network is allocated a limited number of frequencies. These are distributed throughout the cells in their network.
- ▼ Depending on the traffic load and the availability of frequencies, a cell may have one or more frequencies allocated to it.
- ▼ **Interference** is the sum of all signal contributions that are neither noise or the wanted signal. The interference create degradation of signal quality

III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

► Problems about frequencies:

- ❖ **Co-channel interference (C/I)**: Co-channel interference is caused by the use of a frequency close to the exact same frequency.
- ❖ **Adjacent channel interference (C/A)**: Adjacent frequencies (A), that is frequencies shifted 200kHz from the carrier frequency (C), must be avoided in the same cell and preferably in neighboring cells also

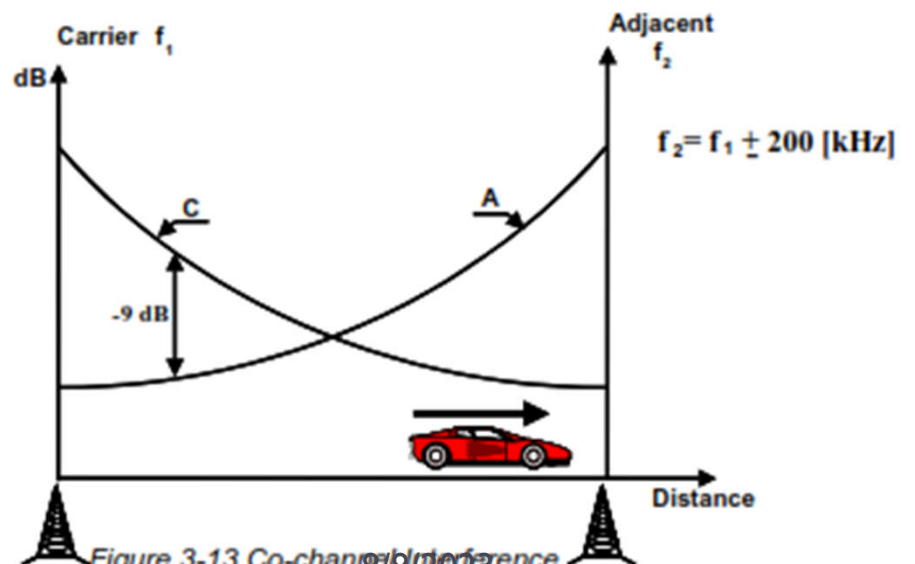
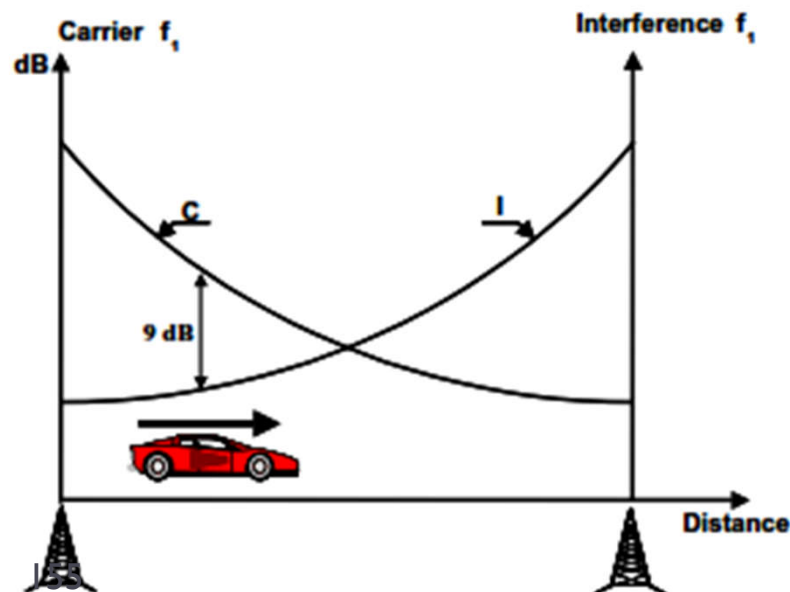


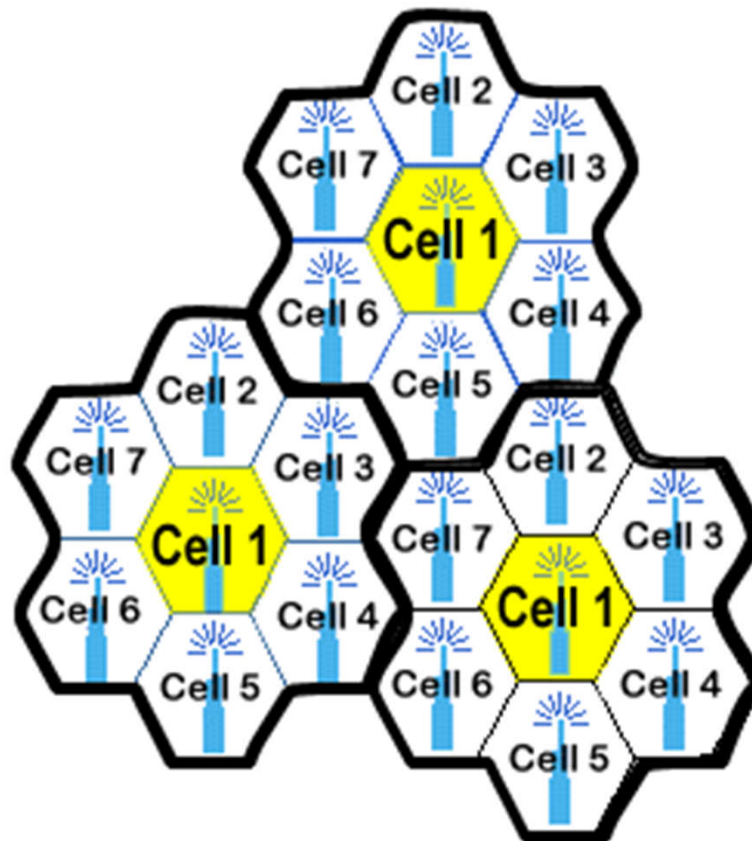
Figure 3-13 Co-channel interference

8/8/2023

III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

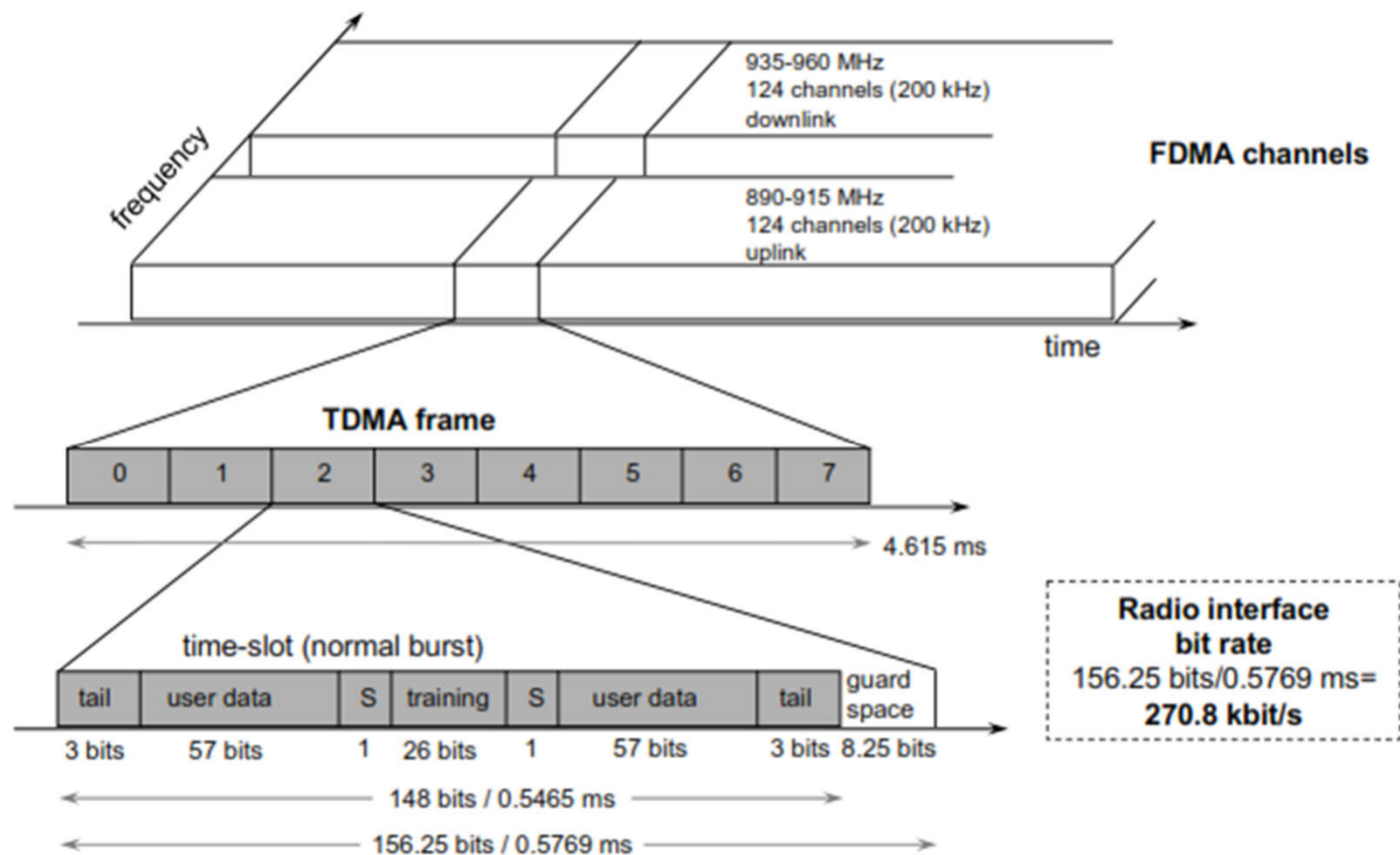
- ▶ **Frequency re-use** is applied to improve quality and capacity.



III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

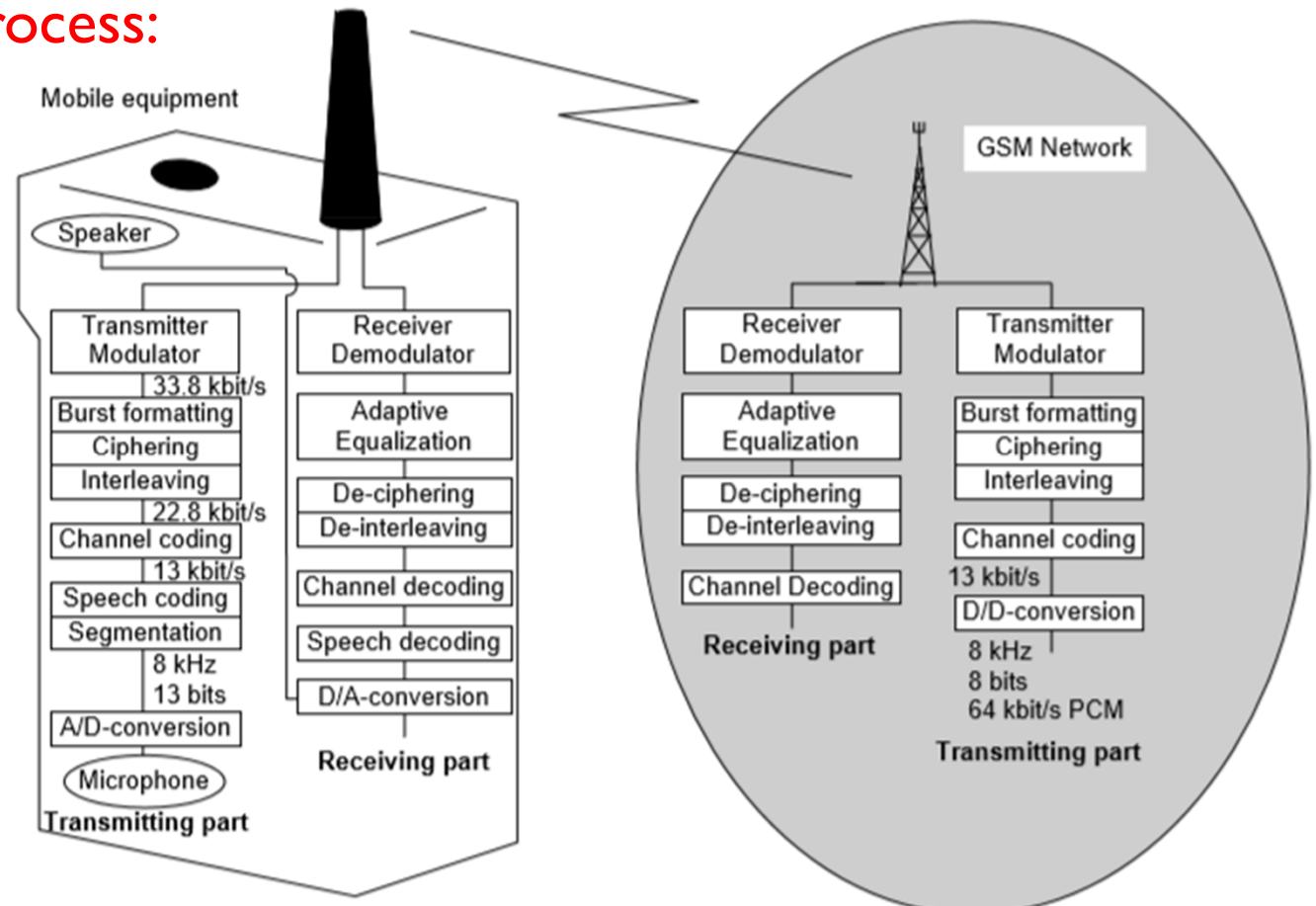
► Information transmission:



III. Wireless Transmission (cont.)

1. Some main characteristics (cont.)

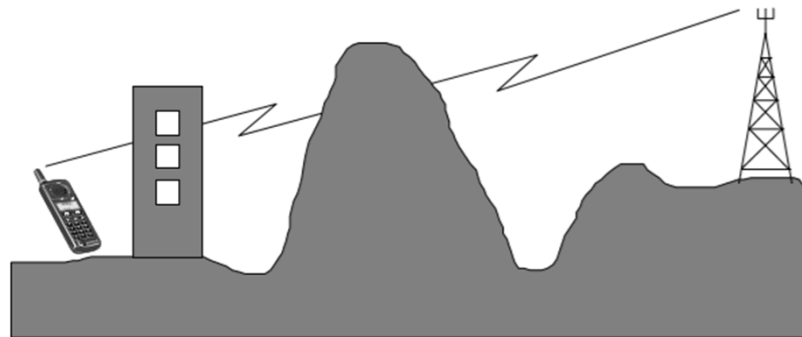
► Transmission process:



III. Wireless Transmission (cont.)

2. Transmission problems

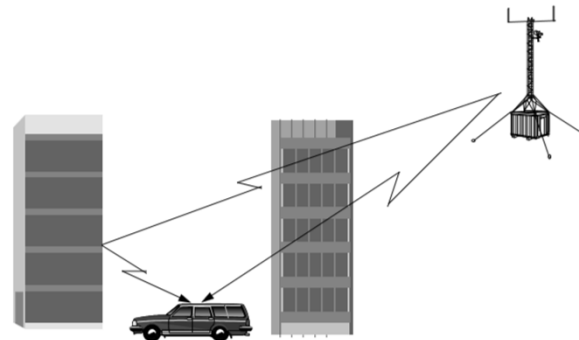
- ▶ **Path loss** occurs when the received signal becomes weaker and weaker due to increasing distance between MS and BTS, even if there are no obstacles between the transmitting (Tx) and receiving (Rx) antenna.
- ▶ **Shadowing** occurs when there are physical obstacles including hills and buildings between the BTS and the MS. The obstacles create a shadowing effect which can decrease the received signal strength



III. Wireless Transmission (cont.)

2. Transmission problems (cont.)

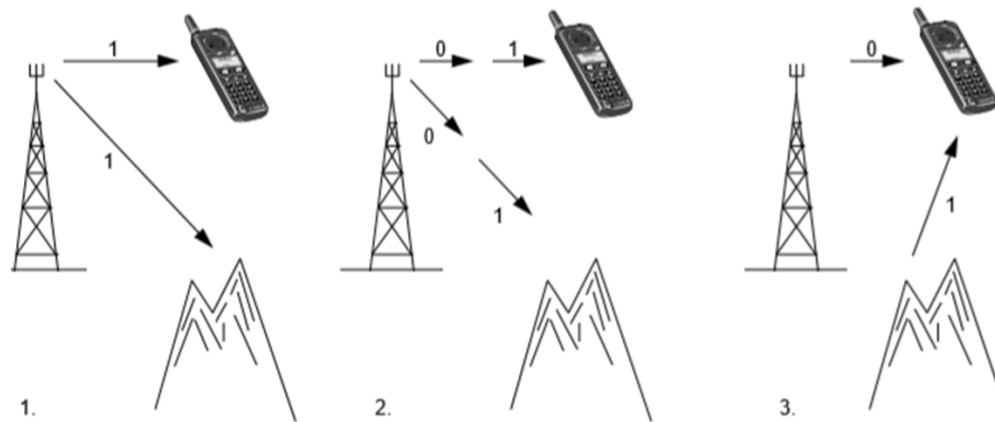
- ▶ **Multipath fading** occurs when there is more than one transmission path to the MS or BTS, and therefore more than one signal arriving at the receiver. This may be due to buildings or mountains, either close to or far from the receiving device.
- ✓ **Rayleigh fading** occurs when a signal takes more than one path between the MS and BTS antennas. In this case, the signal is not received on a line-of-sight path directly from the Tx antenna. Rather, it is reflected off buildings. Rayleigh fading occurs when the obstacles are close to the receiving antenna



III. Wireless Transmission (cont.)

2. Transmission problems (cont.)

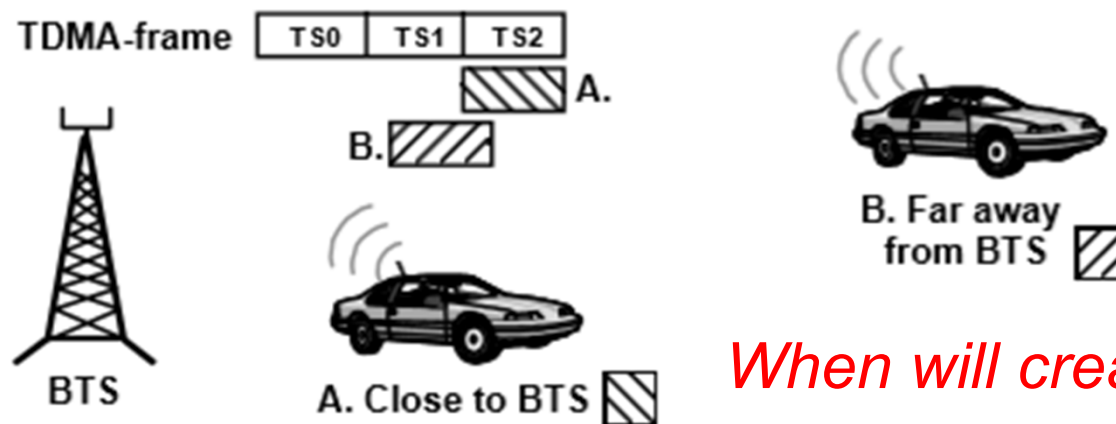
- ✓ **Time dispersion** is another problem relating to multiple paths to the Rx antenna of either an MS or BTS. However, in contrast to Rayleigh fading, the reflected signal comes from an object far away from the Rx antenna
- ✓ Time dispersion causes Inter-Symbol Interference (ISI) where consecutive symbols (bits) interfere with each other making it difficult for the receiver to determine which symbol is the correct one.



III. Wireless Transmission (cont.)

2. Transmission problems (cont.)

- Each MS on a call is allocated a time slot on a TDMA frame. This is an amount of time during which the MS transmits information to the BTS. The information must also arrive at the BTS within that time slot. The **time alignment** problem occurs when part of the information transmitted by an MS does not arrive within the allocated time slot.

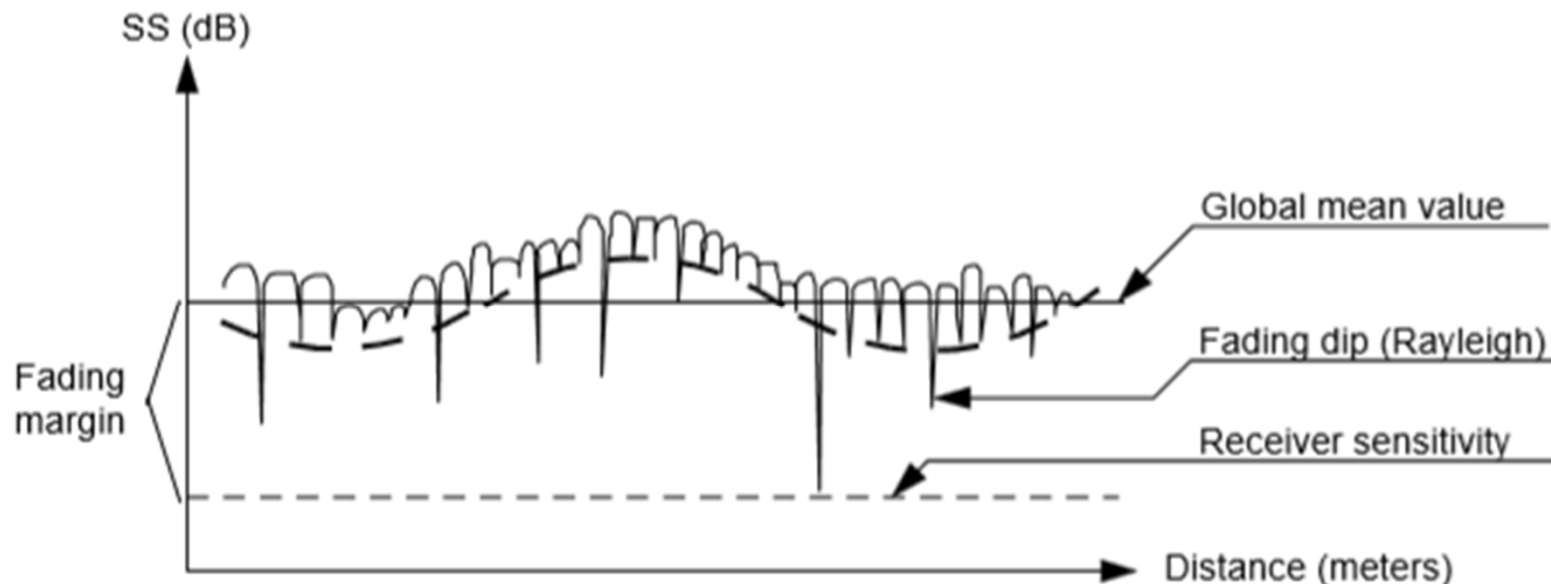


When will create time alignment?

III. Wireless Transmission (cont.)

2. Transmission problems (cont.)

- ▶ The lowest signal strength value required for a specified output is called **receiver sensitivity** level.

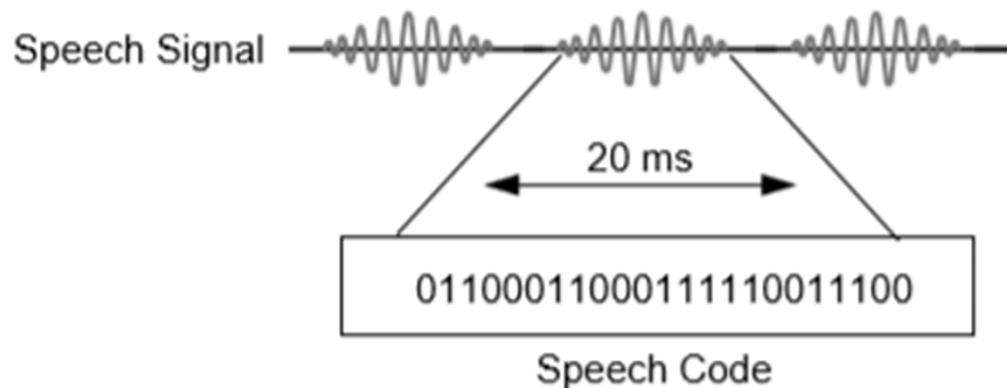


What happened if the global mean is smaller the receiver sensitivity?

III. Wireless Transmission (cont.)

3. Solutions to transmission problems

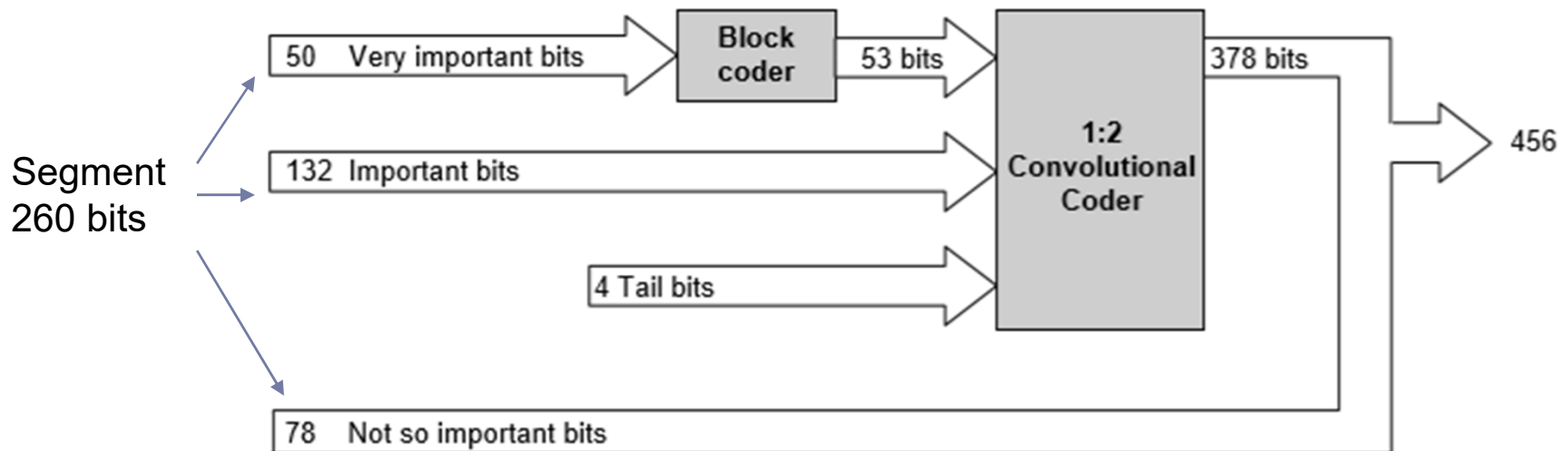
- ▶ **Speech coding:** Instead of using 13 bits per sample as in A/D conversion, GSM speech coding uses 260 bits/20ms. This calculates as $50 \times 260 = 13 \text{ kbits/s}$.



III. Wireless Transmission (cont.)

3. Solutions to transmission problems

- ▶ **Channel coding:** Channel coding is used to detect and correct errors in a received bit stream. Channel coding in GSM uses the 260 bits and split according to their relative importance:
 - Block 1: 50 very important bits
 - Block 2: 132 important bits and
 - Block 3: 78 not so important bits

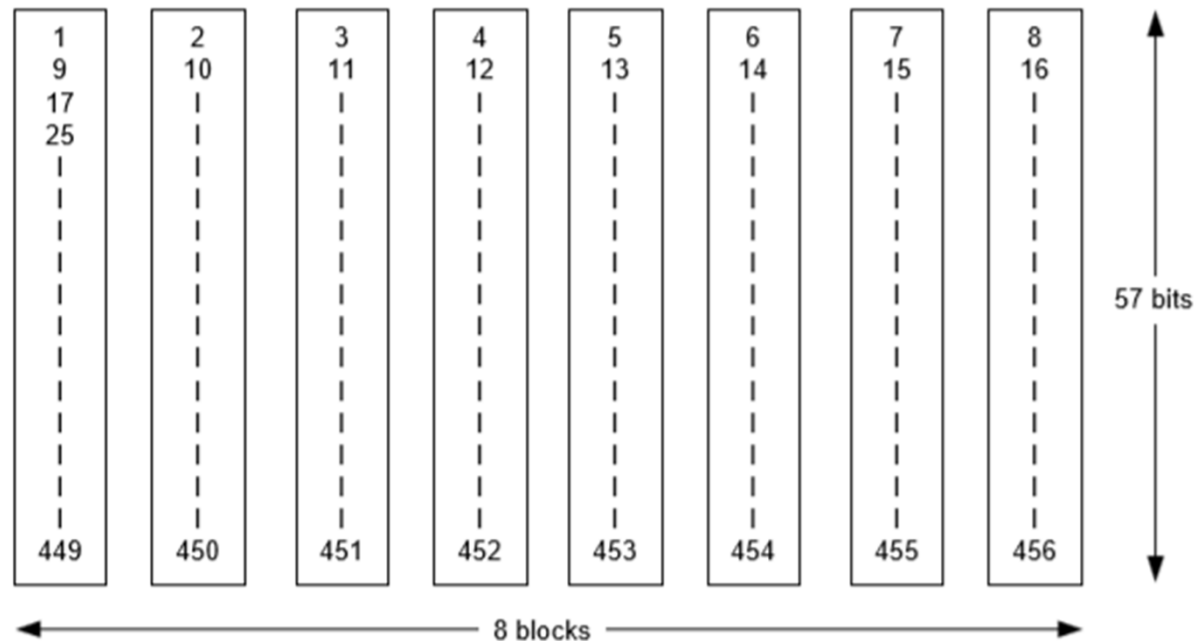


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

- **Interleaving:** There two level of interleaving

- ❖ First interleaving:



If one burst transmission is lost, What is the BER for the entire 20 ms of speech?

III. Wireless Transmission (cont.)

3. Solutions to transmission problems

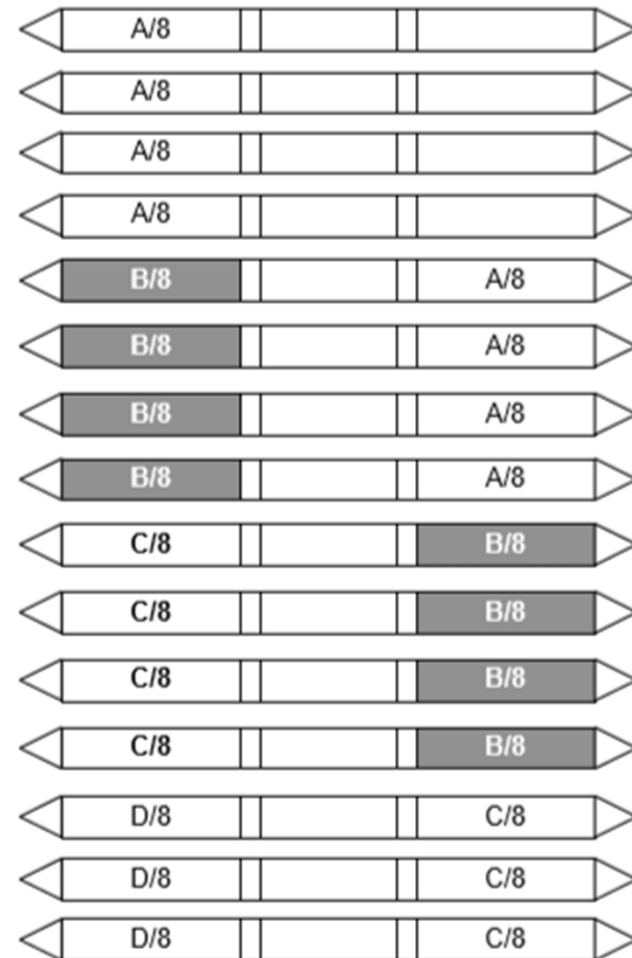
► Interleaving (cont.):

❖ Second interleaving:

A	B	C	D
20 ms speech 456 bits = 8×57	20 ms speech 456 bits = 8×57	20 ms speech 456 bits = 8×57	20 ms speech 456 bits = 8×57

If one burst transmission is lost, what is the BER for the entire 20 ms of speech?

Burst Frames

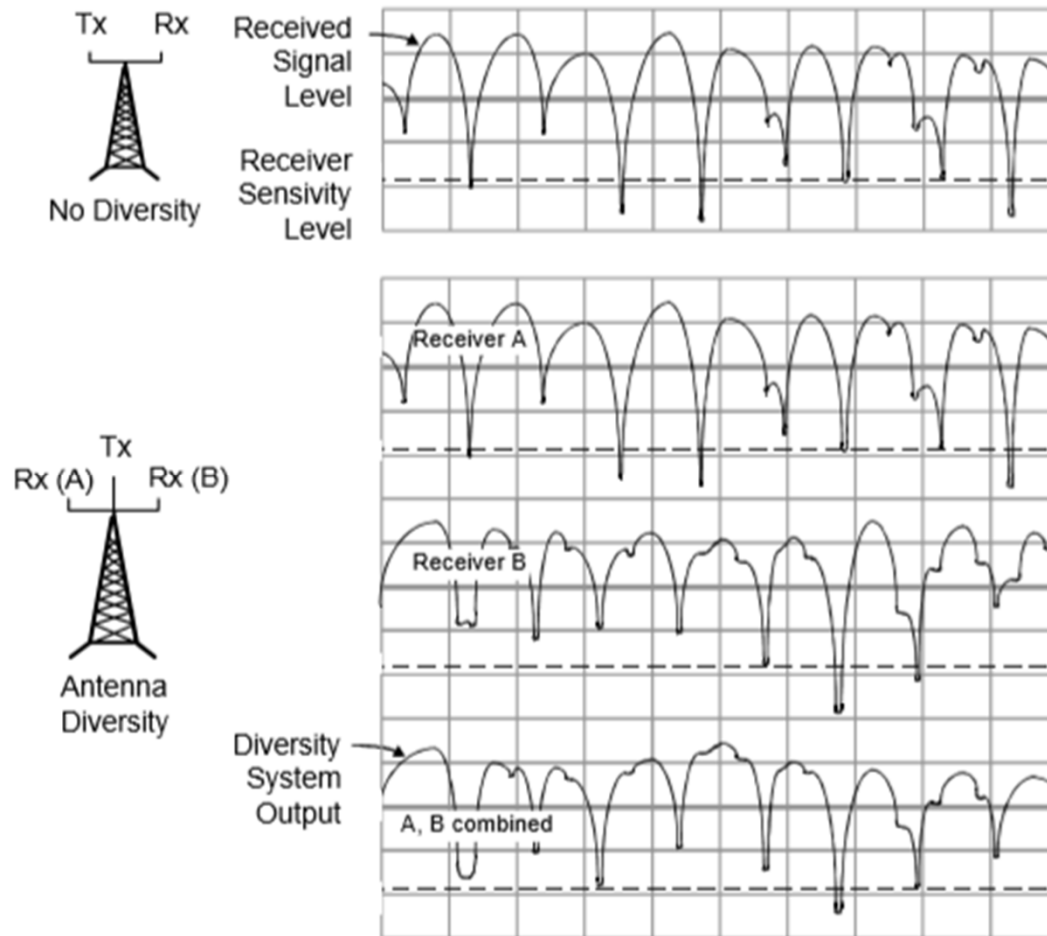


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

- ▶ **Antenna diversity:**

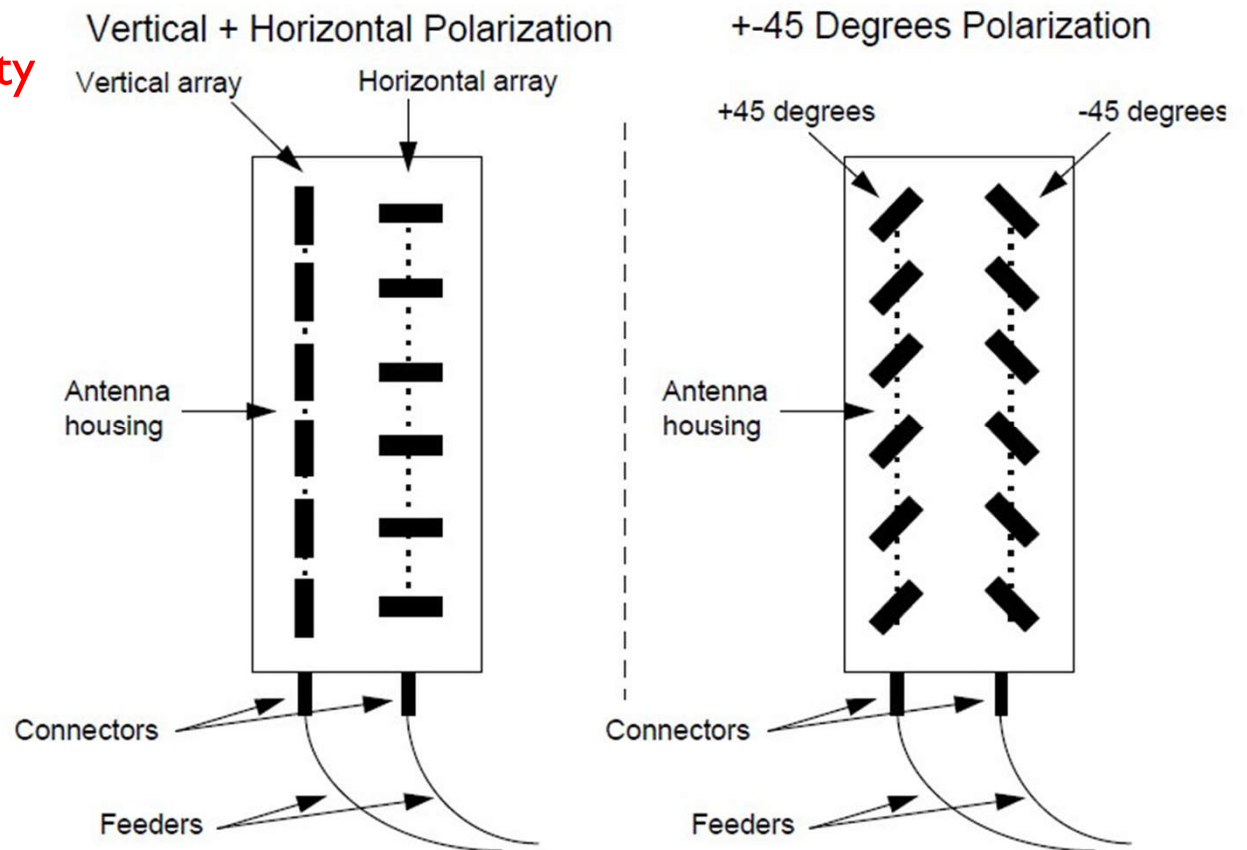
- ❖ **Space diversity:**



III. Wireless Transmission (cont.)

3. Solutions to transmission problems

- ▶ **Antenna diversity:**
 - ❖ **Polarization Diversity**

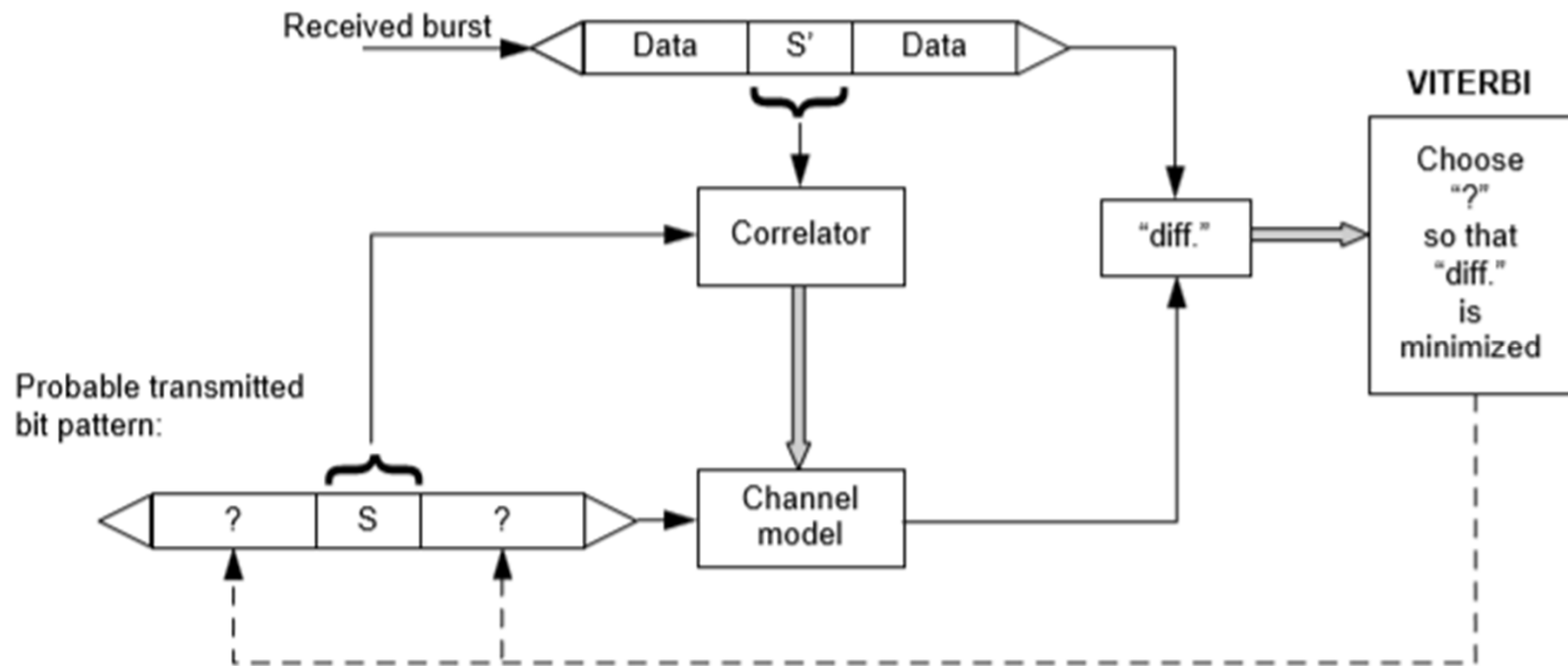


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

► Adaptive equalization:

A set of predefined known bit patterns exist, known as training sequences (S). These are known to the BTS and the MS.

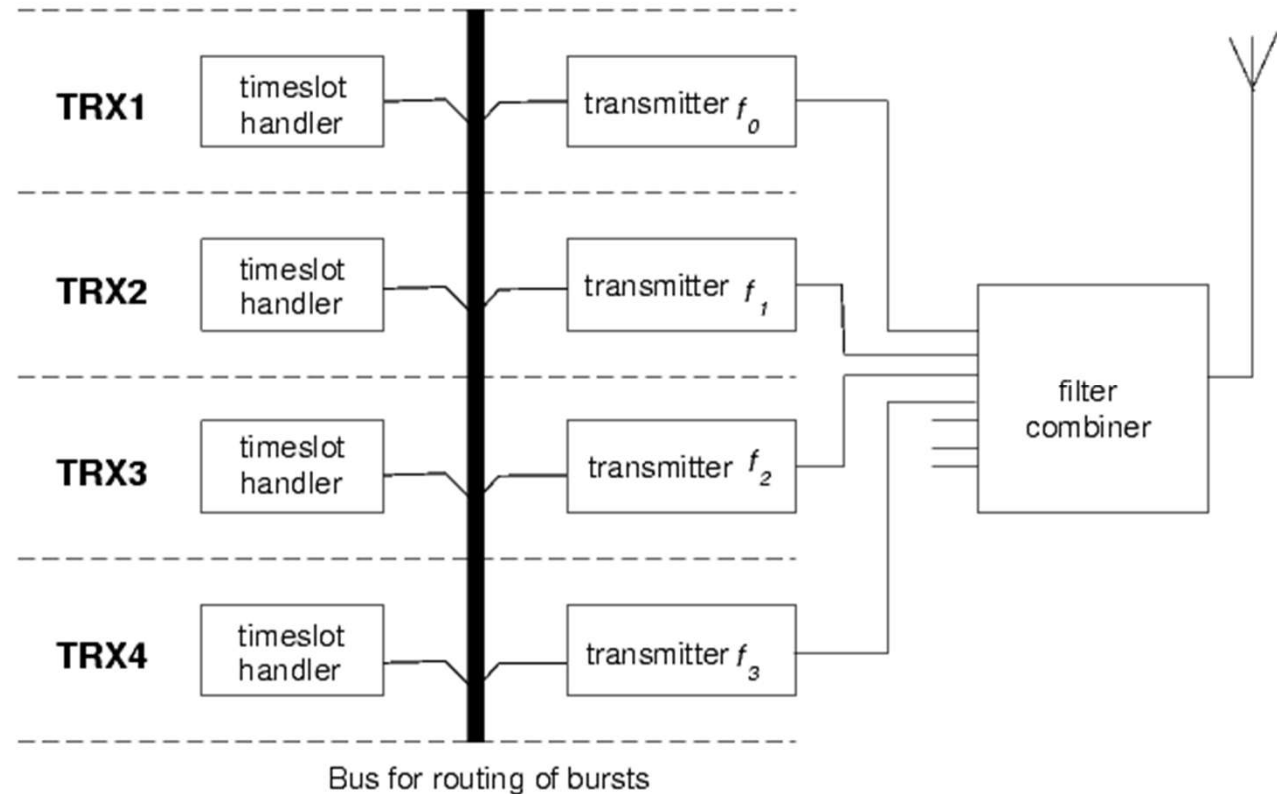


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

► Low frequency hopping (LH):

It is assumed there are 4 carrier frequencies in one BTS and a frame TDM have 8 time slots

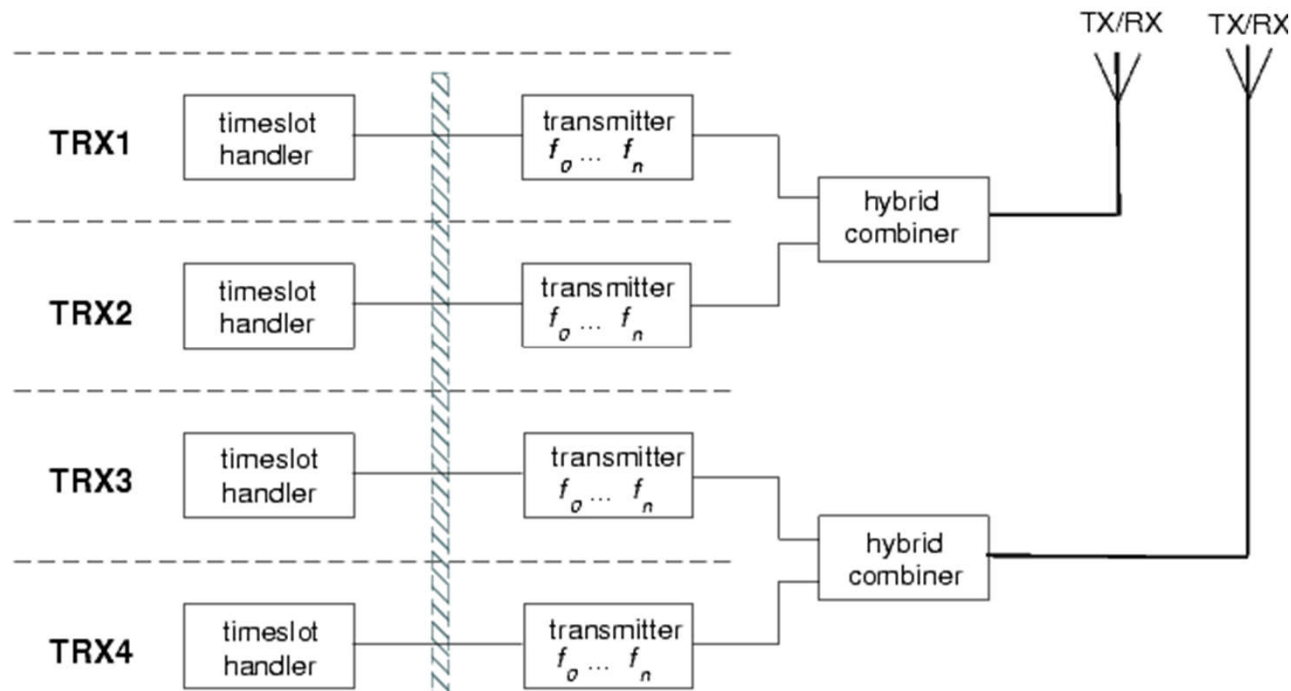


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

► Fast frequency hopping (FH):

It is assumed there are 4 carrier frequencies in one BTS and a frame TDM have 8 time slots

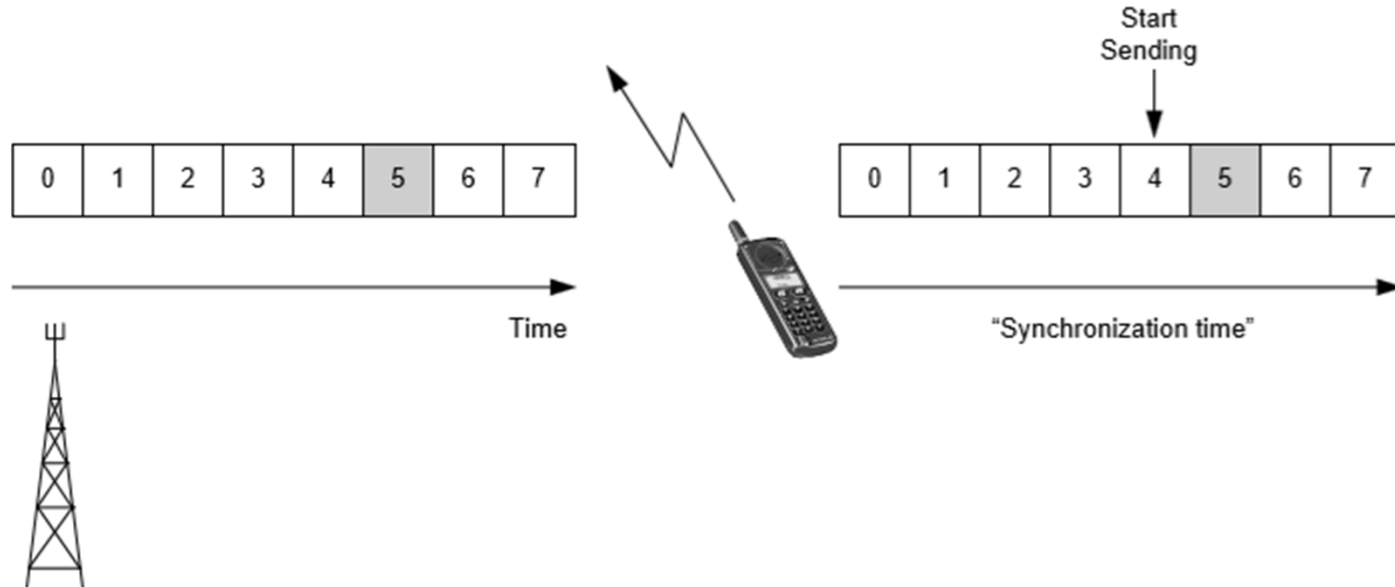


III. Wireless Transmission (cont.)

3. Solutions to transmission problems

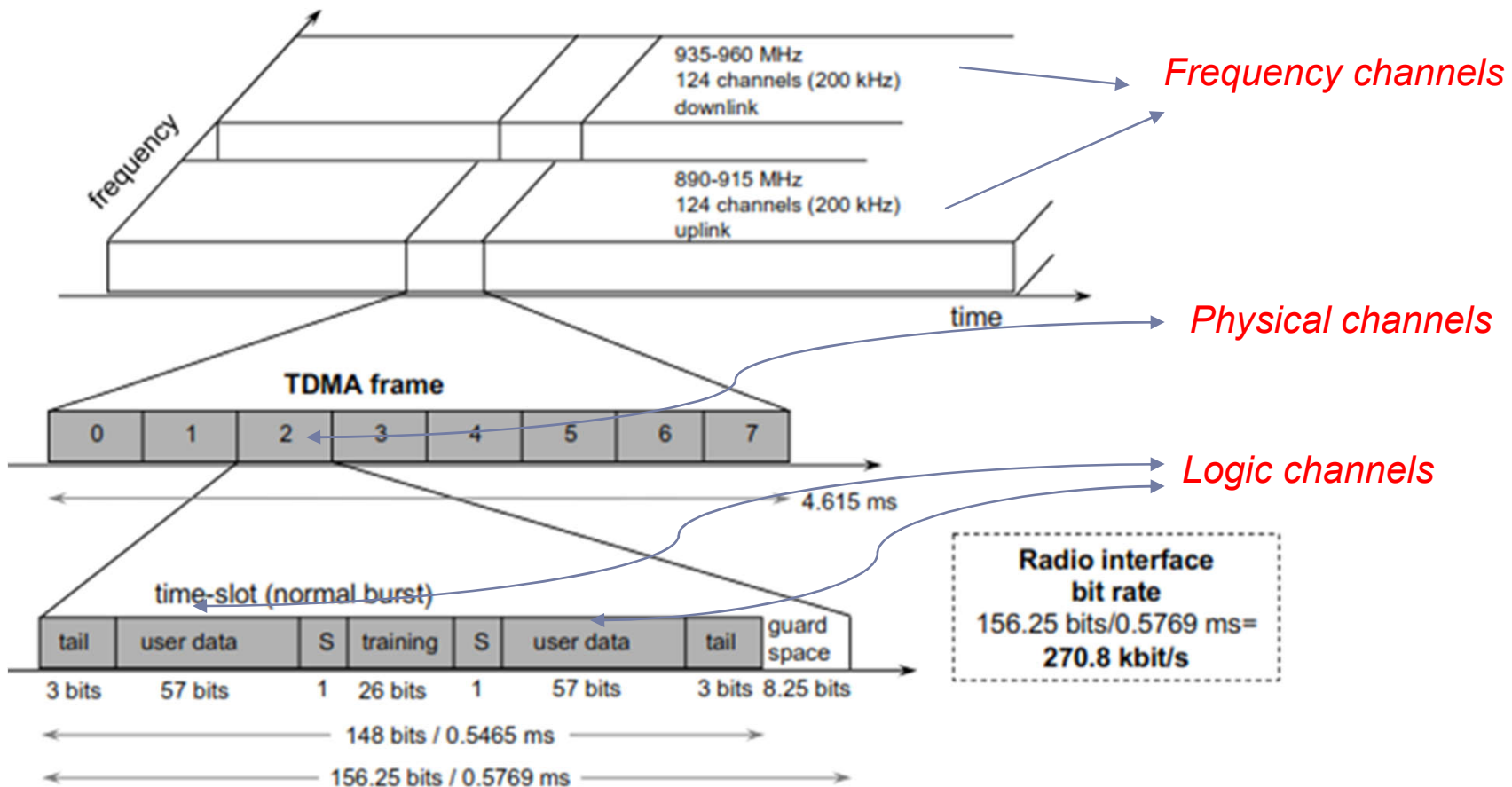
► Timing advance (TA):

In GSM, the timing advance information relates to bit times. Thus, an MS may be instructed to advance its transmission by a certain number of bit times. The maximum in GSM is 63 bit times. This is one of the parameters that limits the GSM cell size to a maximum of 35 km radius



IV. Transmission channels

1. Types of channels:



IV. Transmission channels (cont.)

2. Logical channels:

Broadcast Channels (BCH):

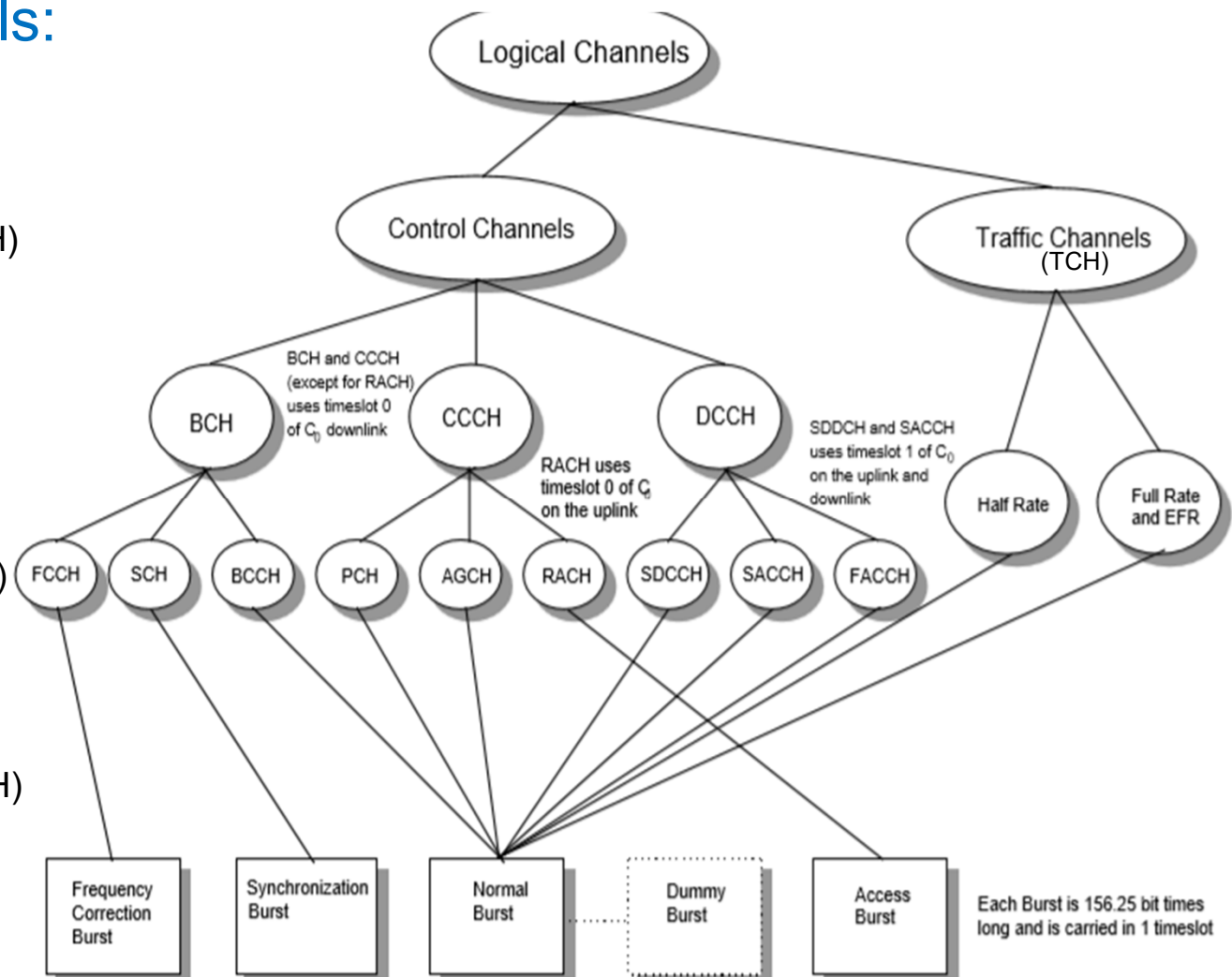
- Frequency Correction Channel (FCCH)
- Synchronization Channel (SCH)
- Broadcast Control Channel (BCCH)

Common Control Channels (CCCH):

- Paging Channel (PCH)
- Random Access Channel (RACH)
- Access Grant Channel (AGCH)

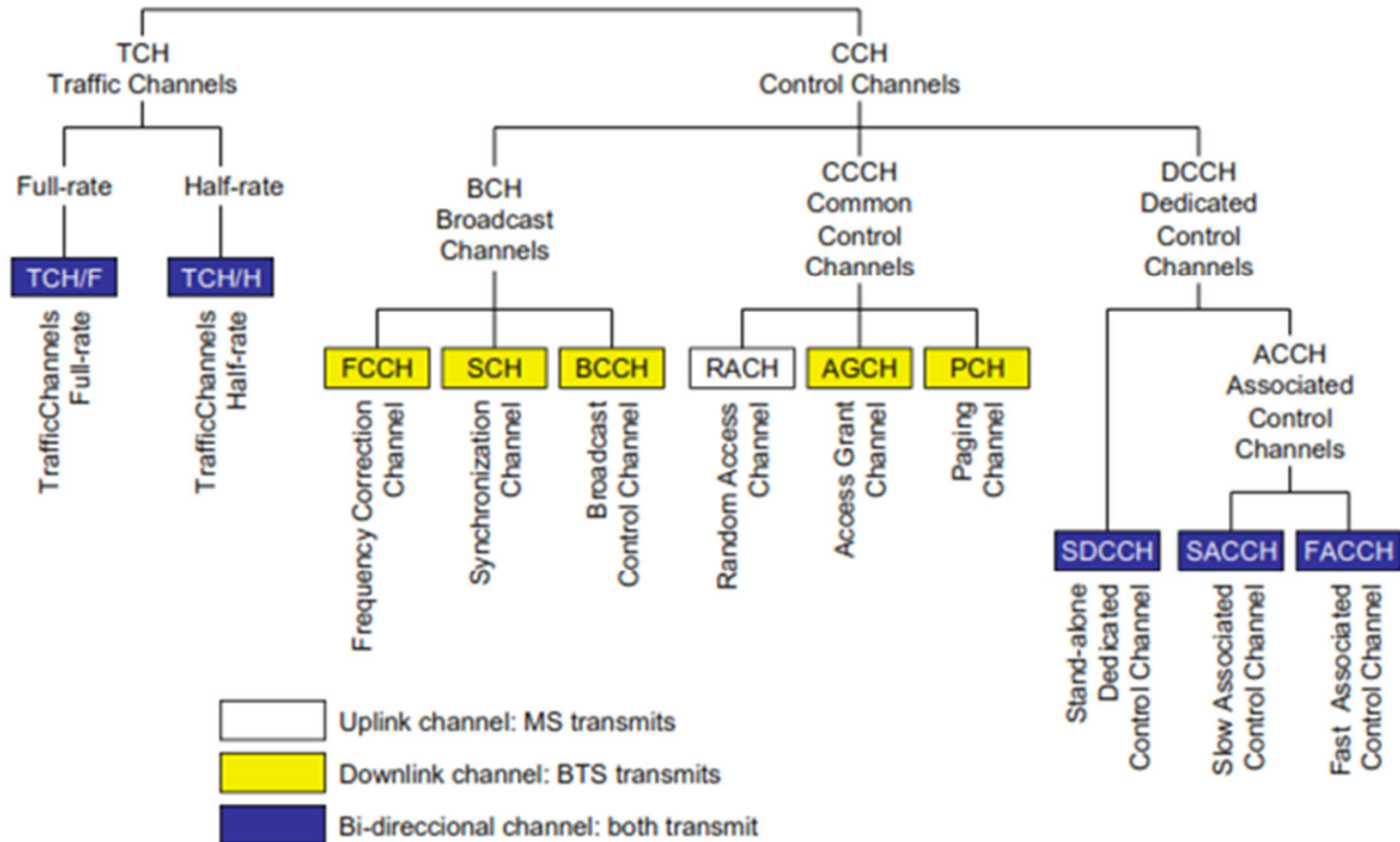
Dedicated Control Channels (DCCH):

- Stand alone Dedicated Control Channel (SDCCH)
- Cell Broadcast Channel (CBCH)
- Slow Associated Control Channel (SACCH)
- Fast Associated Control Channel (FACCH)



IV. Transmission channels (cont.)

2. Logical channels:



IV. Transmission channels (cont.)

2. Logical channels:

Broadcast Channels (BCH):

- Frequency Correction Channel (FCCH)
- Synchronization Channel (SCH)
- Broadcast Control Channel (BCCH)

Common Control Channels (CCCH):

- Paging Channel (PCH)
- Random Access Channel (RACH)
- Access Grant Channel (AGCH)

Dedicated Control Channels (DCCH):

- Stand alone Dedicated Control Channel (SDCCH)
- Cell Broadcast Channel (CBCH)
- Slow Associated Control Channel (SACCH)
- Fast Associated Control Channel (FACCH)

Burst Type	Purpose	Used by	Contents
Normal	Used to carry information on traffic and control channels	BCCH, PCH, AGCH, SDCCH, CBCH, SACCH, FACCH, TCH	• Two blocks of 57 bits each for traffic • Training sequence (26 bits) • Steal flags (1 bit each) to indicate that FACCH has temporarily stolen 57 bits • Tail bits (always 000) • Guard period: 8.25 bit durations
Frequency Correction	Used for frequency synchronization of the mobile	FCCH	• 142 frequency correction bits • Tail bits • Guard period: 8.25 bit durations
Synchronization	Used for frame synchronization of the mobile	SCH	• Two blocks of 39 bits for TDMA frame structure information • 64 synchronization bits • Tail bits • Guard period: 8.25 bit durations
Access	Used for random and handover access	RACH	• 41 synchronization bits • 36 bits of access information • Tail bits • Guard period: 68.25 bit durations. A longer GP is used because it's the first transmission from the mobile - no timing advance information is available
Dummy	Used when no other channel requires a burst to be sent and carries no information		• Pattern is identical to normal burst, but carries no information

IV. Transmission channels (cont.)

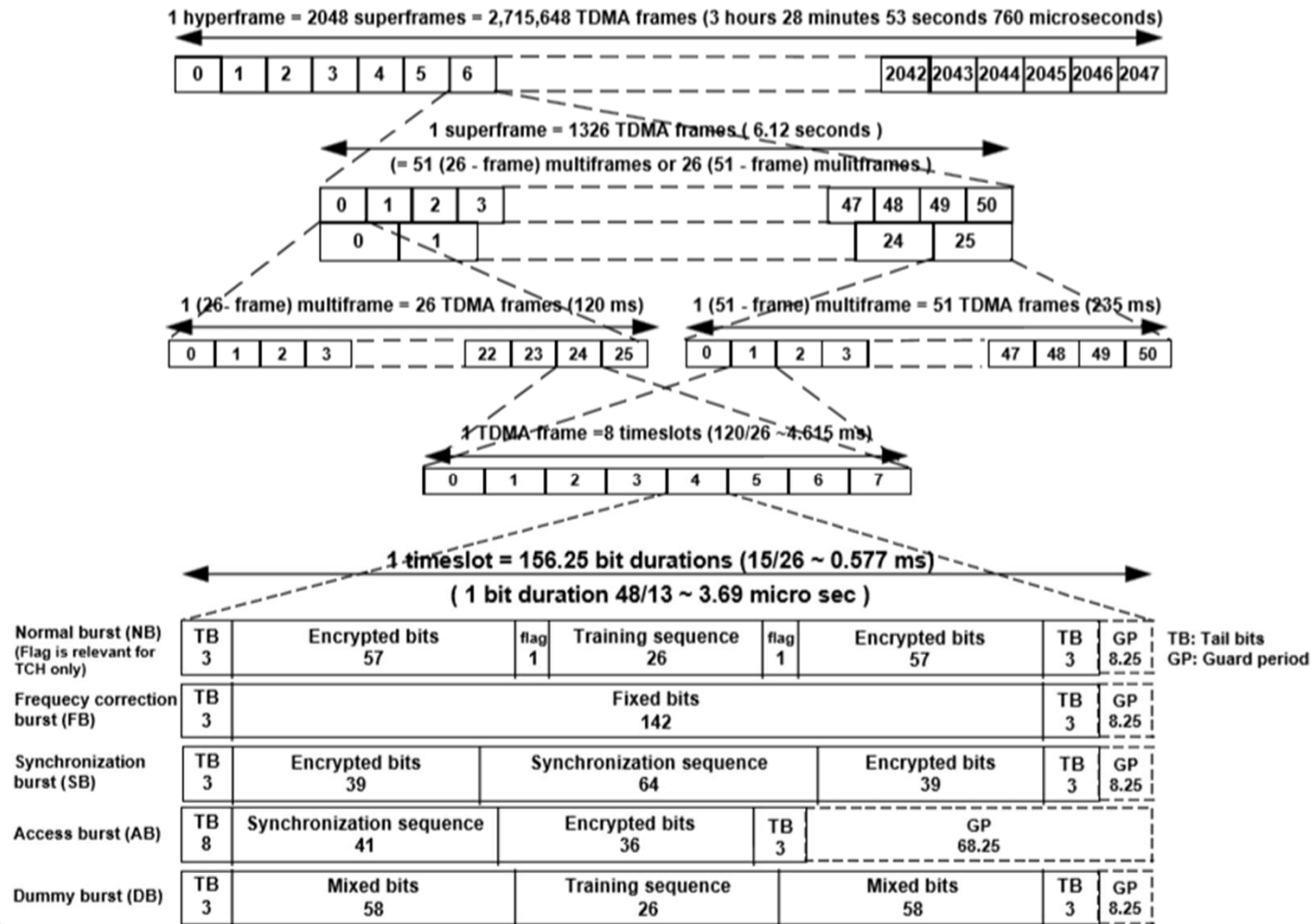
2. Logical channels:

Channel		Direction	Application	Allocation
TCH Traffic Channels	TCH/H	BTS ↔ MS	User data	Allocated by network on demand by MS
	TCH/F			
BCH Broadcast Channels	FCCH	BTS → MS	Carrier synchronization	Permanent
	SCH		Frame synchronisation	
	BCCH		General network information Cell information (present and adjacent)	
CCCH Common Control Channels	RACH	BTS ← MS	Request SDCCH for signalling Request TCH for handover	Multiple access with slotted Aloha contention between MS
	AGCH	BTS → MS	Confirmation of SDCCH or TCH request	Permanent
	PCH		Alert MS to a call originated in the network	
DCCH Dedicated Control Channels	SDCCH	BTS ↔ MS	Registration / location updating Call control procedures	Allocated by network on demand
	SACCH		Control information between MS and BTS during the progress of a call or call set up	Associated to a specific TCH or SDCCH
	FACCH		Exchange of time critical control information during the progress of a call	Allocated by network or MS (*)

(*) Fast allocation by setting S bit; bits are stolen from TCH

IV. Transmission channels (cont.)

2. Logical channels:



IV. Transmission channels (cont.)

3. Mapping the logical channels in physical channels:

Legend:
B: BCH
C: CCCH
D: DCCH
T: TCH

		Time slot							
		0	1	2	3	4	5	6	7
Carrier Frequency	0	B,C	T	D	T	T	T	T	T
	1	T	T	D	T	T	T	T	T
	2	T	T	T	T	T	T	T	T
	3	T	T	T	T	T	T	T	T

(This model is applied for GSM Ericsson)

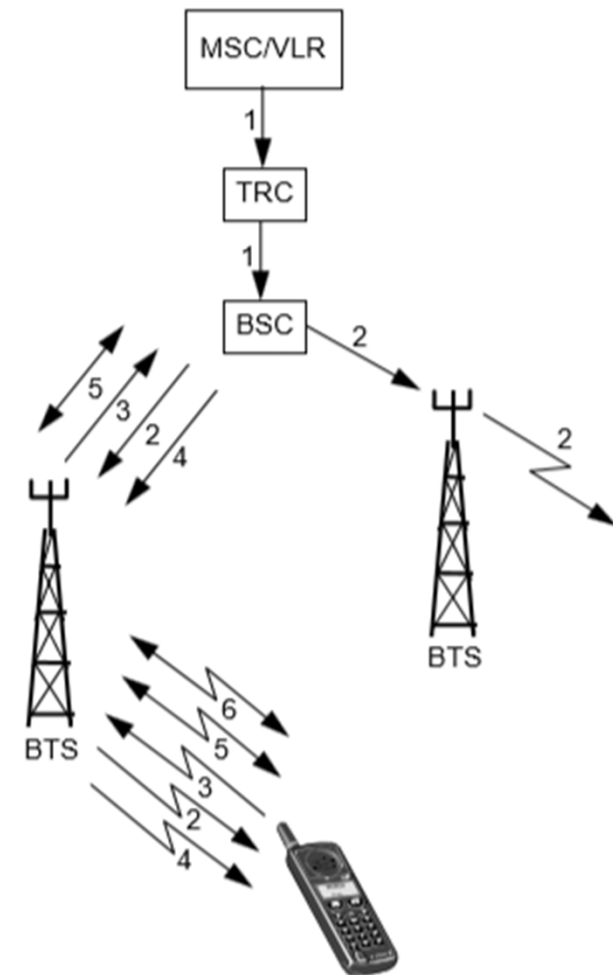
For each site, we usually use 2 or 3 slots for signaling

IV. Transmission channels (cont.)

4. Sample traffic case - call to an MS:

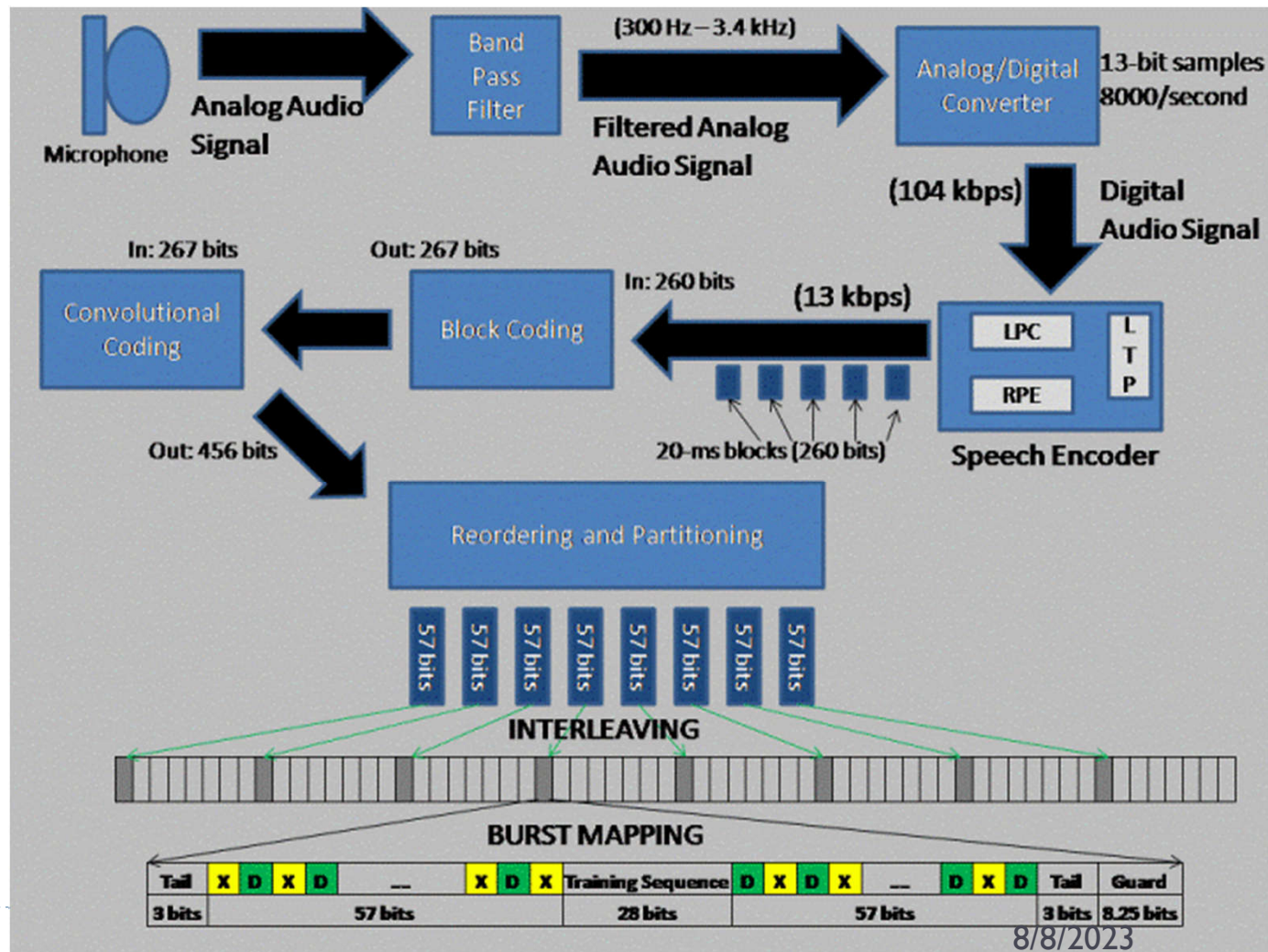
1. The MSC/VLR knows which LA the MS is located in. A paging message is sent to the BSCs controlling the LA.
2. The BSCs distribute the paging message to the BTSs in the desired LA. The BTSs transmit the message over the air interface using **PCH**.
3. When the MS detects a PCH identifying itself, it sends a request for a signaling channel using **RACH**.
4. The BSC uses **AGCH** to inform the MS of the signaling channel (**SDCCH** and **SACCH**) to use.
5. **SDCCH** and **SACCH** are used for call set-up. A **TCH** is allocated and the **SDCCH** is released.
6. The MS and BTS switch to the identified **TCH** frequency and time slot. The MS rings. If the subscriber answers, the connection is established. During the call, the radio connection is maintained by information sent and received by the MS using **SACCH**.

Hint: Transcoder Controller (TRC)



IV. Transmission channels (cont.)

5. Process from voice signal to data channels:



V. Cell planning

1. Concepts:

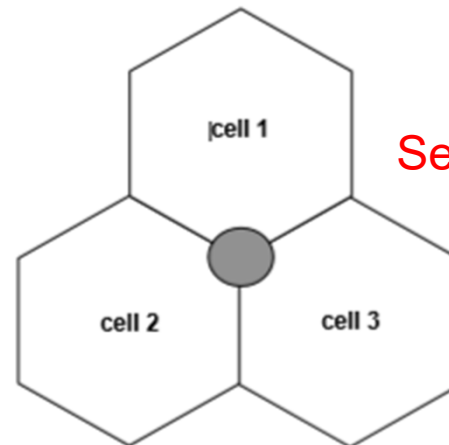
► There are two main types of cell:

- ✓ **Omni directional cell:** An omni directional cell (or omnicell) is served by a BTS with an antenna which transmits equally in all directions (360 degrees).
- ✓ **Sector cell:** A sector cell is the area of coverage from an antenna which transmits in a given direction only. For example, this may be equal to 120 degrees or 180 degrees of an equivalent omni directional cell.

Omni directional cell



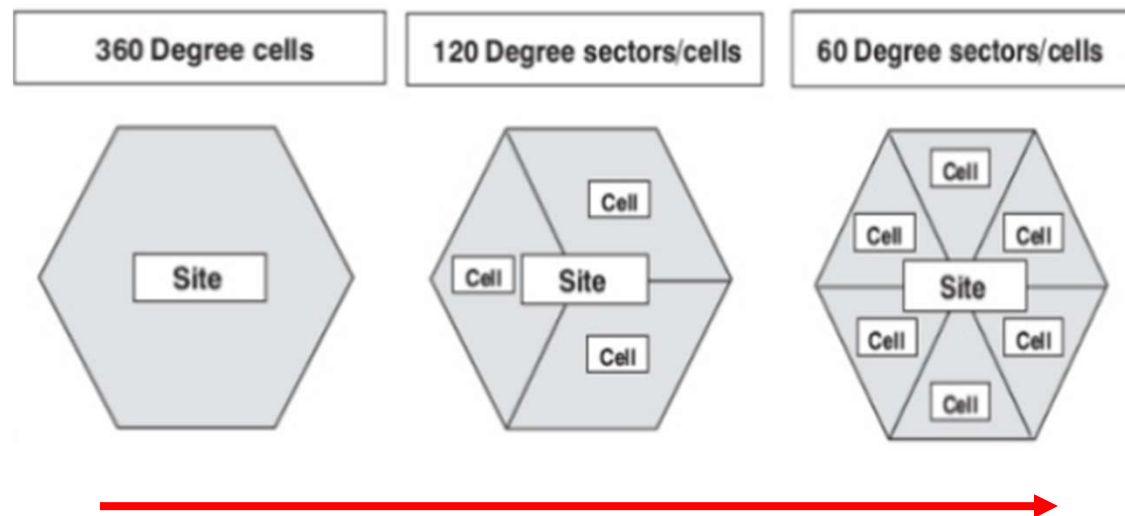
Sector cell:



V. Cell planning

1. Concepts:

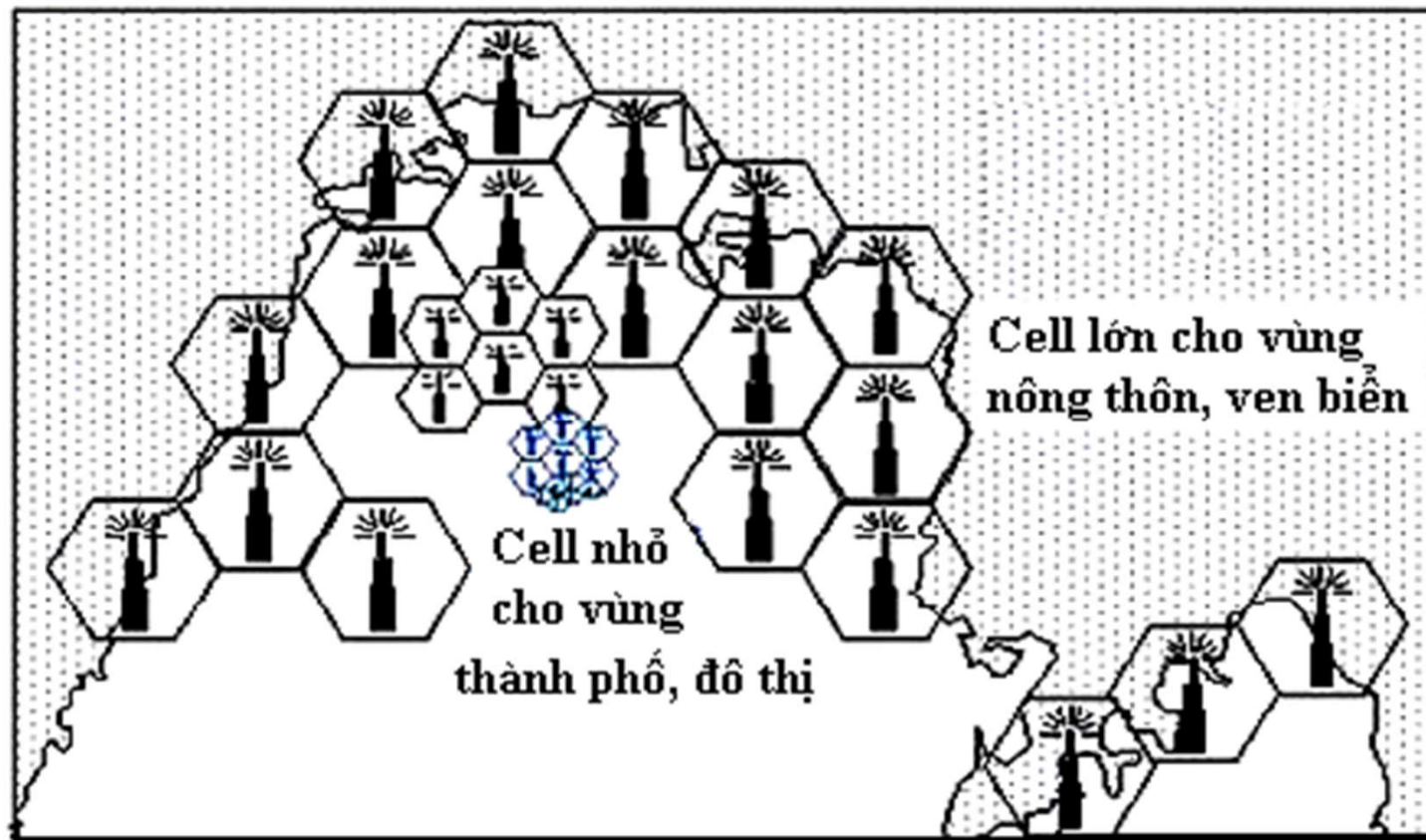
- ▶ **Cell splitting** is the process of subdividing a congested cell into smaller cells. In that way, each smaller cell has reduced antenna height and reduced transmitter power. It also increases the capacity of a cellular system since number of times channels are reused increases.



V. Cell planning

1. Concepts:

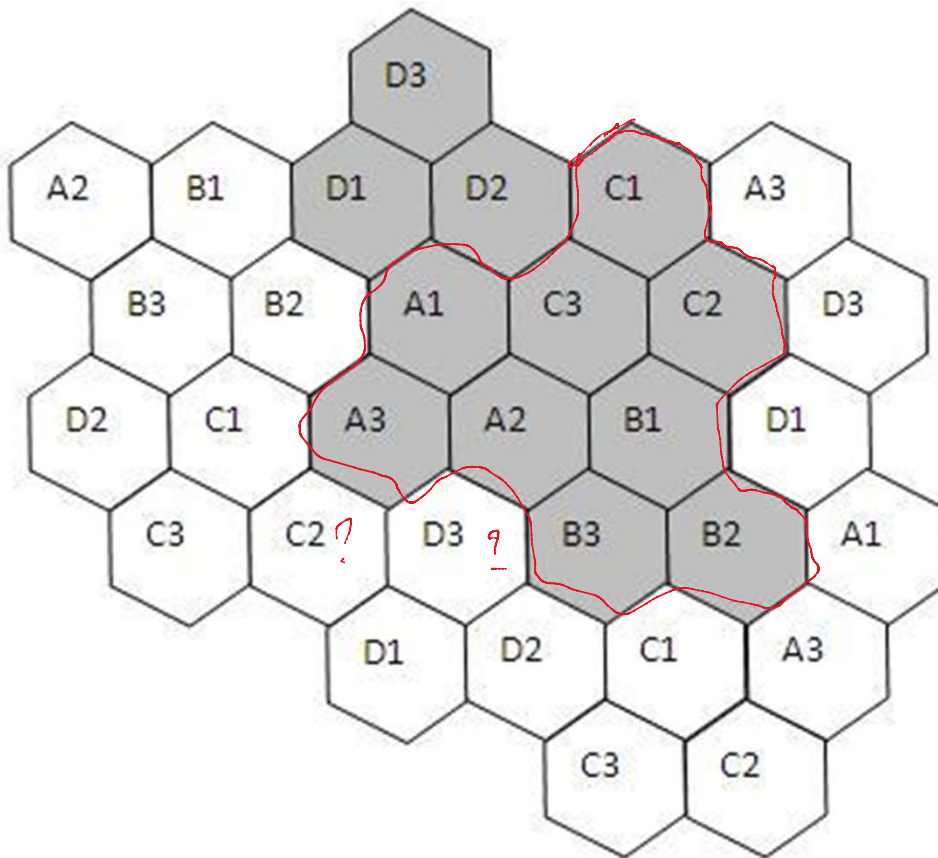
► Cell splitting (cont.):



V. Cell planning

1. Concepts:

► Configuration for frequency re-use:



Groups of frequencies can be placed together into patterns of cells called **clusters**.

A cluster is a group of cells in which all available frequencies have been used once and only once.

This model using **cluster configuration** (re-use pattern) = $4 \text{ (sites)}/12 \text{ (cells)}$

V. Cell planning

1. Concepts:

► Distribution of carrier frequencies:

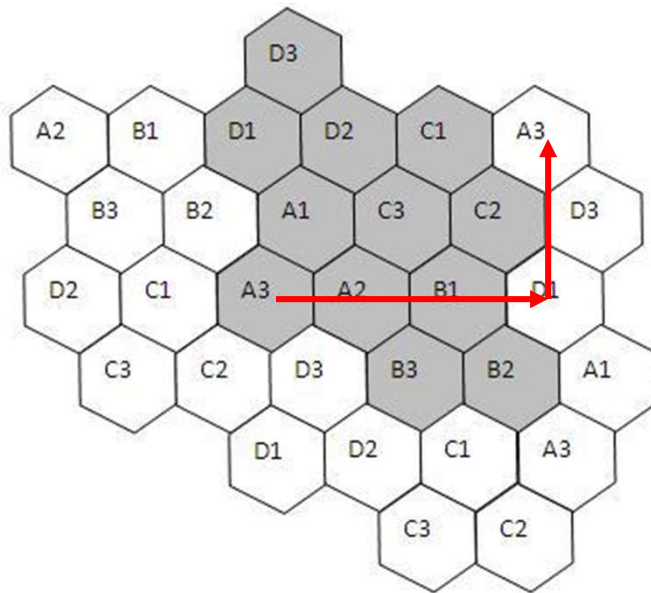
Below is an example of how a network operator could divide 24 available frequencies (1-24) into a 3/9 cell pattern:

Frequency group	A1	B1	C1	A2	B2	C2	A3	B3	C3
Channels	1	2	3	4	5	6	7	8	9
	10	11	12	13	14	15	16	17	18
	19	20	21	22	23	24			

V. Cell planning

1. Concepts:

► Configuration for frequency re-use:



Some cluster configuration: 3/9, 4/12 and 7/21

Assume that R is cell radius, N is the number of cells in a cluster and D is re-use distance as the distance between centers of the nearest **co-channel** cells

For this model:

$$D = \sqrt{(3\sqrt{3}R)^2 + (3R)^2} = 6R$$

General formulations for D :

$$D = R\sqrt{3N}$$

V. Cell planning

1. Concepts:

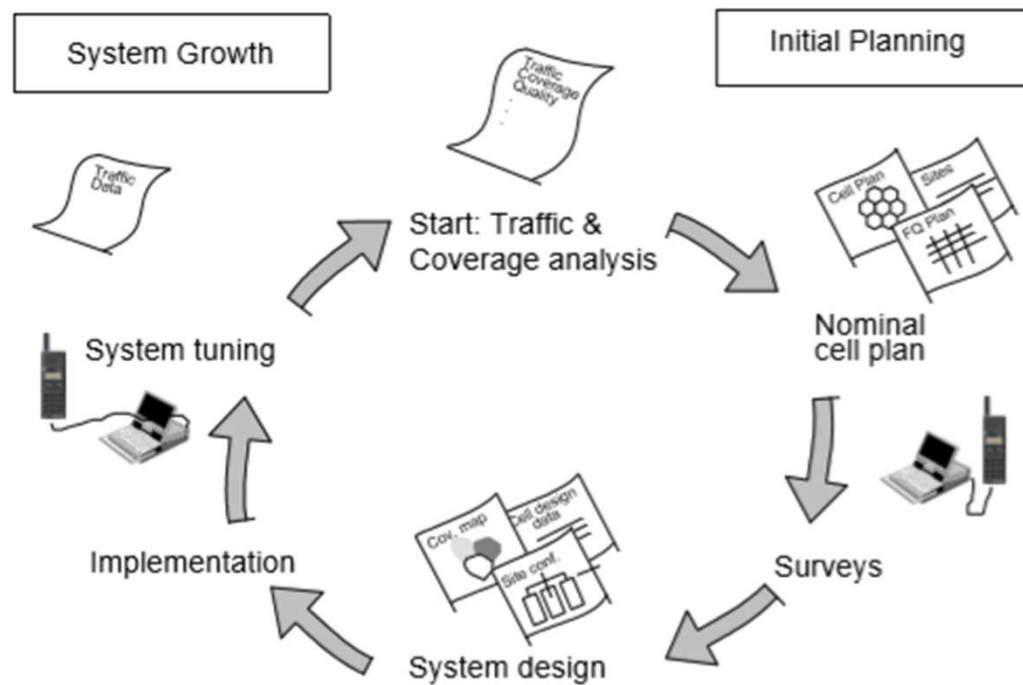
► Calculate C/I_{dB} :

Calculate C/I of GSM network using the re-use pattern 4/12. It is known that the radius of cells is 2 Km and path loss formulation is $L = \left(\frac{4\pi d}{\lambda}\right)^2$

V. Cell planning

2. Cell planning:

► Cell planning process:

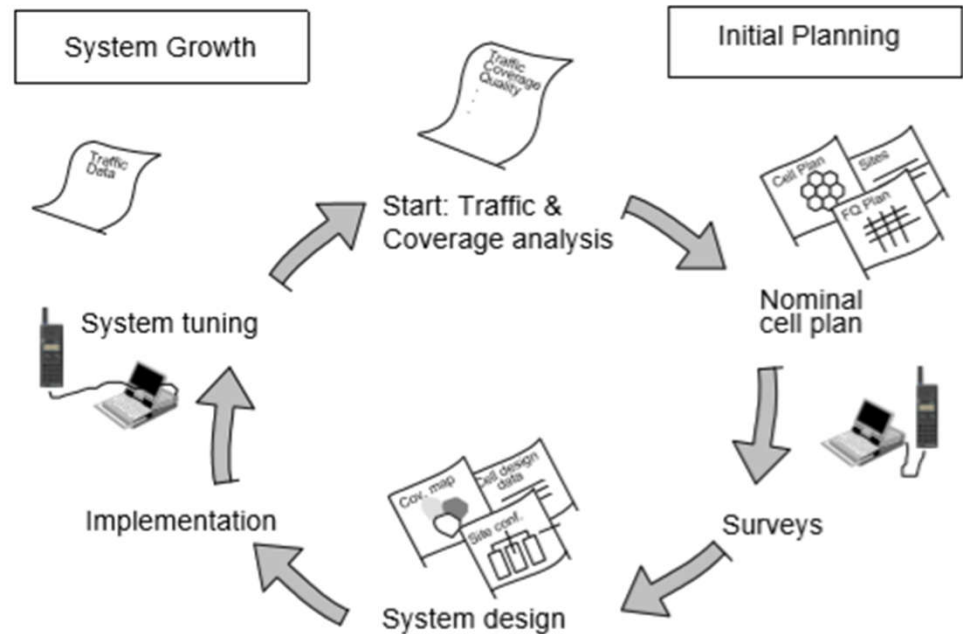


Student is required to read Section 10. Cell Planning in
“Reference [2]. Ericsson_GSM System Survey”

V. Cell planning

2. Cell planning:

► Cell planning process:



Course learning outcomes	Planning steps
CLO3 (d1). Identify input / output design problems with constraints on industry standards, efficiency, environmental and human safety.	Step 1. Traffic & coverage analysis Step 2. Nominal cell plan Step 3. Survey
CLO4 (d2). Design block diagram of the system, define the input / output parameters between the blocks	Step 4. System design

V. Cell planning

2. Cell planning:

- ▶ Step 1. Traffic and coverage analysis

V. Cell planning

2. Cell planning:

- ▶ Step 2. Nominal cell planning

V. Cell planning

2. Cell planning:

- ▶ Step 3. Survey

V. Cell planning

2. Cell planning:

- ▶ Step 4. System design

V. Cell planning

2. Cell planning:

- ▶ Step 5. Implement

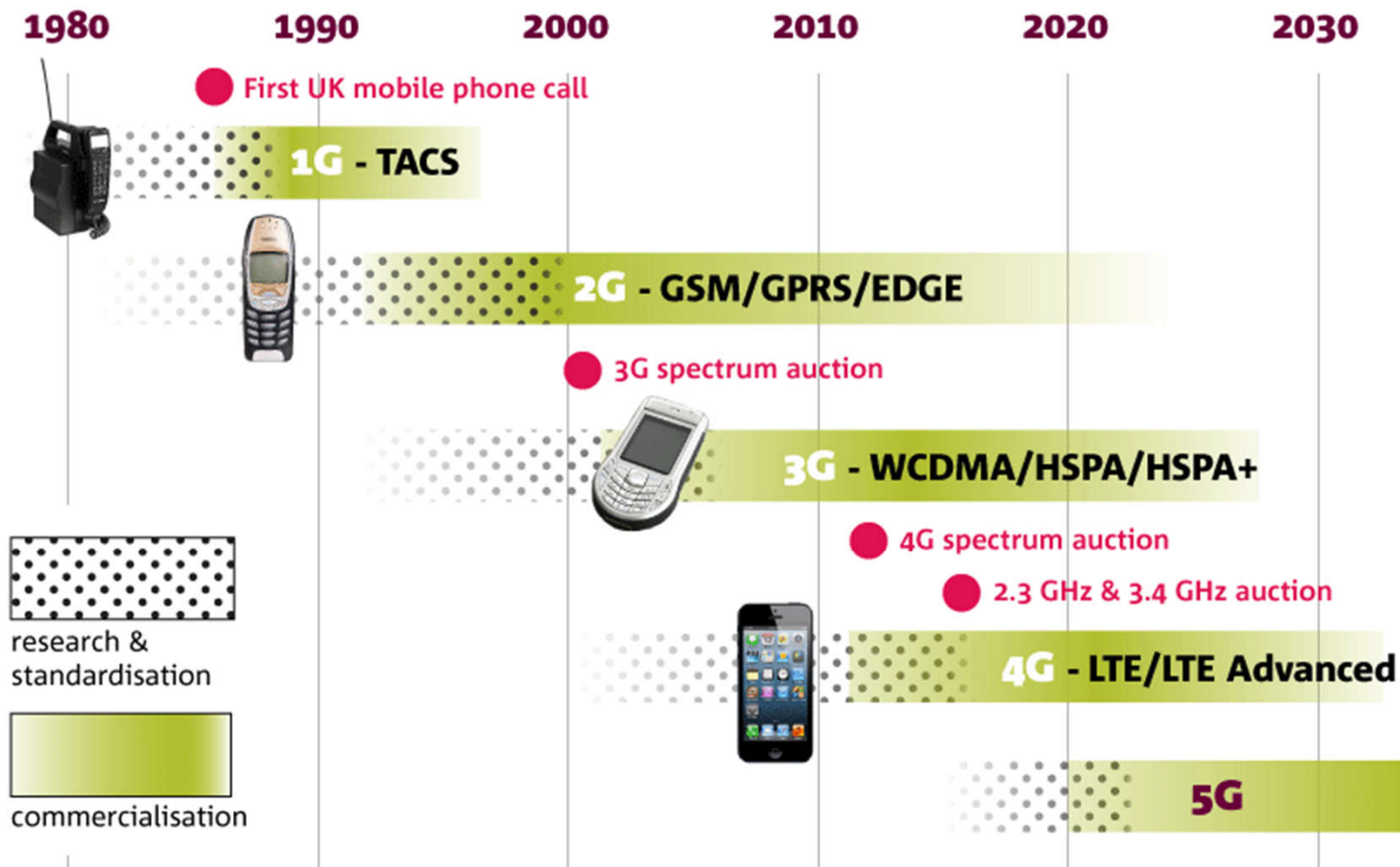
V. Cell planning

2. Cell planning:

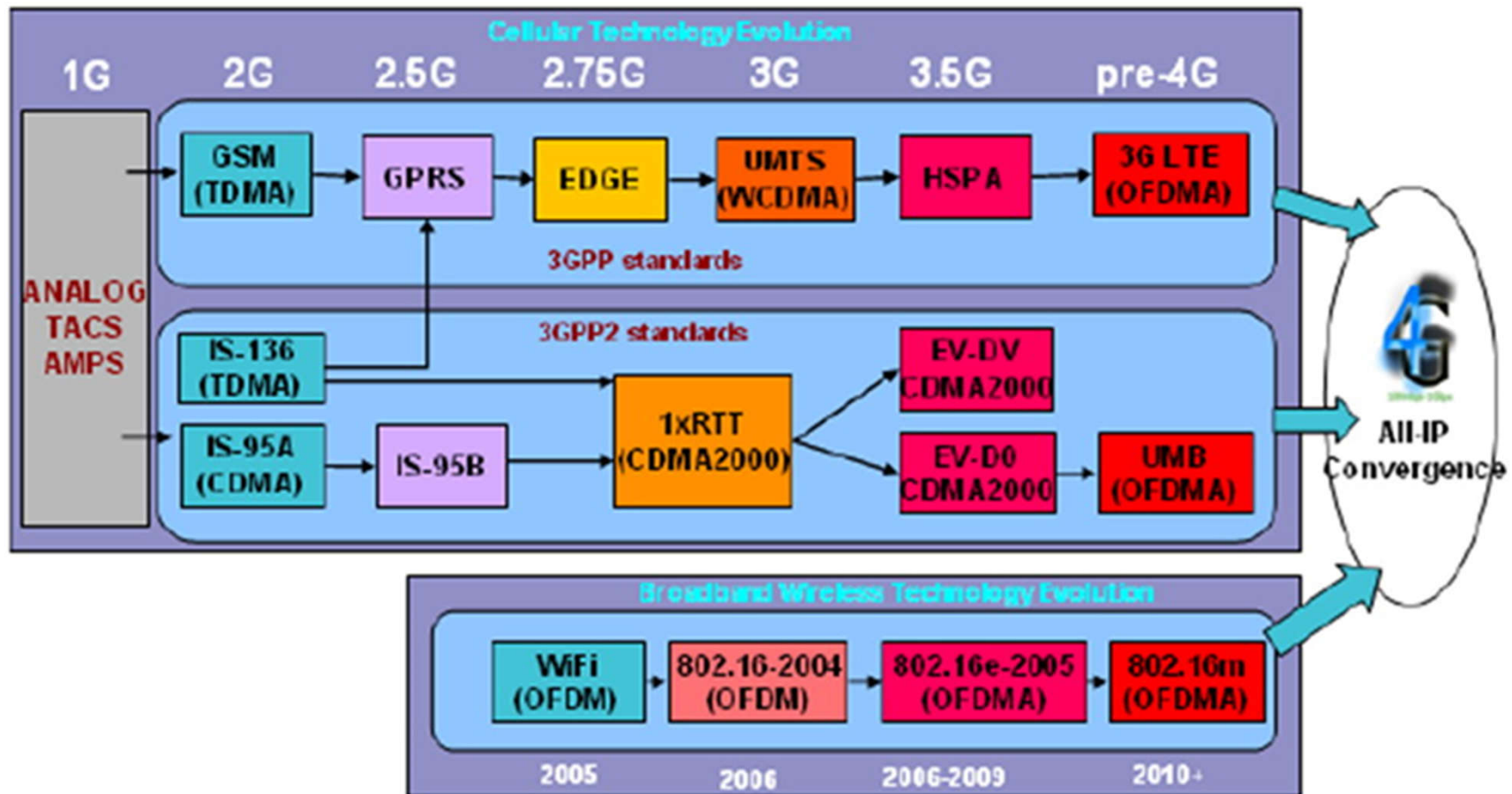
- ▶ Step 6. System turning

VI. Evolution of mobile networks

Evolution of mobile phone communications



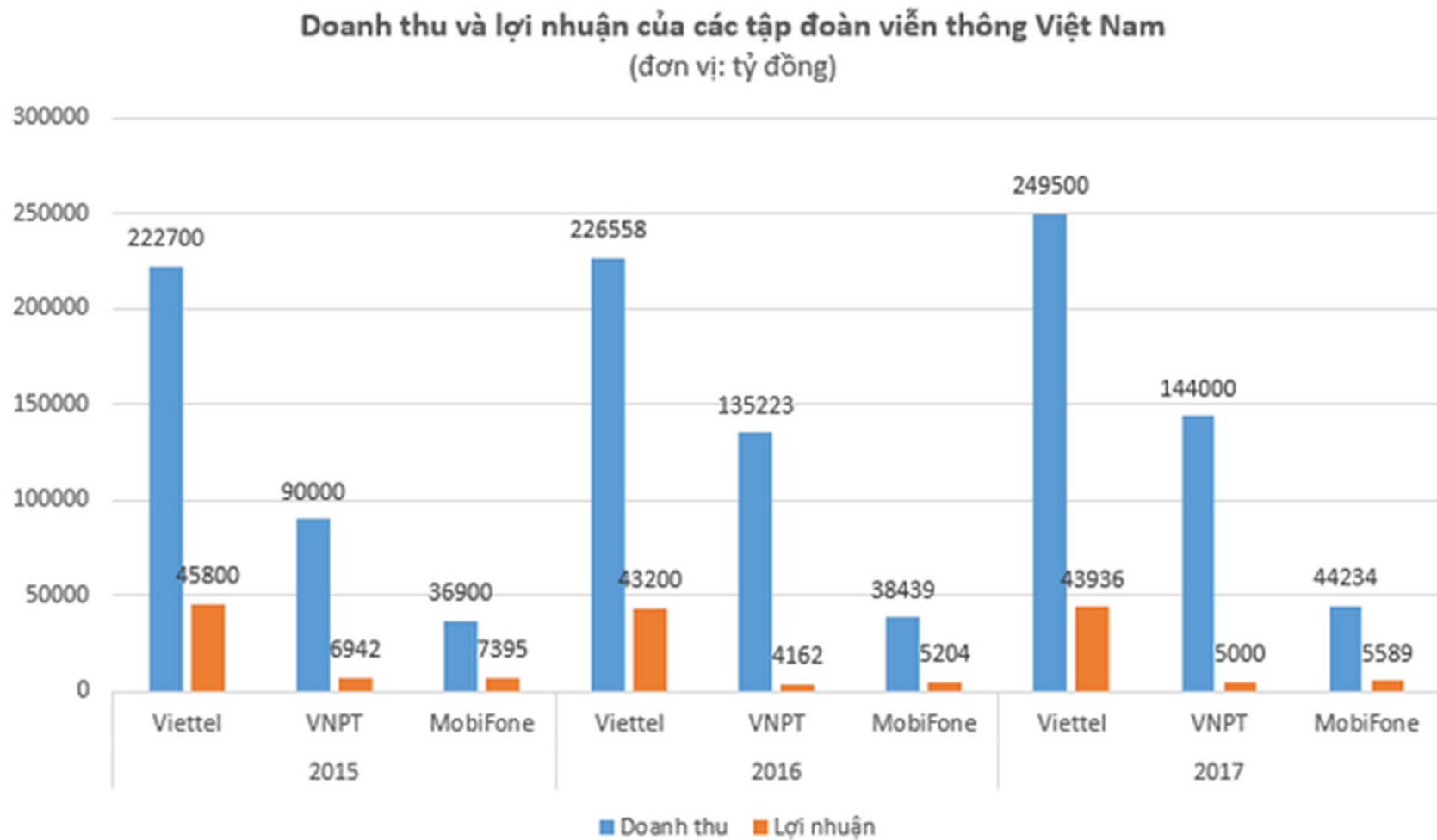
VI. Evolution of mobile networks (cont.)



VI. Evolution of mobile networks (cont.)

1G	2G	3G	4G
			
Mạng không dây thế hệ thứ 1	Mạng không dây thế hệ thứ 2	Mạng không dây thế hệ thứ 3	Mạng không dây thế hệ thứ 4
• Dịch vụ thoại không dây • Giao thức analog	• Cải tiến chất lượng thoại, năng tầm phủ sóng và cung cấp dịch vụ dữ liệu • Chuẩn digital đầu tiên (GSM, CDMA)	• Tối ưu hóa dịch vụ dữ liệu, triển khai dịch vụ băng thông rộng, cho phép sử dụng các ứng dụng video, đồ họa và thoại...	• Cải tiến của 3G về khả năng phục vụ, tốc độ và trải nghiệm băng thông rộng
Tốc độ			
2.4 kbps	64 kbps	2000 kbps	100,000 kbps

VI. Evolution of mobile networks (cont.)



Problems

1. Name mobile network technologies what you know.
2. Present general structure of GSM network and explain function of each modules.
3. Define the SS7 function modules in GSM network.
4. Explain the concepts in mobile network: cell, location area (LA), MSC service area, PLMN.
5. Explain the concepts in mobile network: Idle, active and detached, roaming, international roaming, location update, paging, handover.
6. Define characteristic parameters of GSM900.
7. What is a duplex channel?
8. What is the separation between two consecutive carriers?

Problems

9. Name mobile network technologies what you know.
10. How many slots are used for signalling in a site BTS?
11. Define frequency channel, physical channel and logic channel in mobile network.
12. Present normal burst structure of a logic channel.
13. Present the general diagram of a mobile station (MS).
14. Present the general diagram of a BTS.
15. Describe main transmission problems in mobile network.
16. Define transmission problems are called Rayleigh fading and time dispersion in mobile network.
17. Define solutions for transmission problems in mobile network.
18. Explain the advantage of using interleaving technique.
19. Explain the advantage of using antenna diversity technique.

Problems

20. Explain the advantage of using Viterbi equalization technique.
21. Explain the advantage of timing advance (TA) technique.
21. Present basic structure of logical channels in mobile networks.
22. Describe a traffic case call to MS.
23. Define types of cell in a mobile network and method for splitting cell.
24. What is a cluster?
25. Define the formulation for calculating the distance between co-channel cells.
26. What types of data will be collected in Step 1. Traffic & coverage analysis
27. What types of problems will be done in Step 2. Nominal cell plan
28. What types of data will be collected in Step 3. Survey
29. What types of data will be collected in Step 4. System design

Consider a hexagonal cell with an area of 60 square kilometers. Calculate the frequency reuse distance D if the cellular system is deployed using a 7 cell reuse pattern.

Consider a hexagonal cell with an area of 50 square kilometers. Calculate the frequency reuse distance D if the cellular system is deployed using a 4 cell reuse pattern.

A certain urban has an area of 2,300 square kilometers and is covered by a GSM cellular system using a reuse pattern of 4/12. The population density of this city is two thousand residents per square kilometer. Given that the cellular system is allocated an uplink bandwidth of 20 MHz and a downlink bandwidth of 25 MHz. A full duplex channel bandwidth occupies 200 kHz. Assume the system's QoS is 2%. In addition, a user makes 1.3 calls with a call-holding time of 1.5 minutes, and the out calling is 20% in a busy hour. Assume that each cell in the system gives 3 time slots for signalling and control. Design the cell planning for this system.

A certain urban has an area of 2,350 square kilometers and is covered by a GSM 900 cellular system using a reuse pattern of 4/9. The population density of this city is two thousand residents per square kilometer. Given that the cellular system is allocated an uplink bandwidth of 20 MHz and a downlink bandwidth of 20 MHz. A full duplex channel bandwidth occupies 200 kHz. Assume the system's QoS is 0.015. In addition, a user makes 1.8 calls with a call-holding time of 1.5 minutes and the out calling is 15% in a busy hour. Assume that each cell in the system gives 2 time slots for signalling and control. Design the cell planning for this system.